Name:____________

AS Chemistry Unit 2

Once you have completed a paper and have marked it, fill out the table below, using the grade boundaries provided to add your grade. Then list which questions you did well and which questions you struggled with. For the questions your struggled with, review why and at the reasons. The most common reasons are

- Not understanding basic concepts (e.g. simple definitions)

- Calculation errors (forgetting to square numbers, etc.)

- Not reading questions carefully

- Not answering the full question (describe vs. explain)

- Laboratory/practical setup mistakes

For these questions it is also useful add notes to the question to help you next time and take a photo of these questions. Use these then to guide further revision.

Year	Score /80	Grade	Good Questions	Bad Questions	Reasons
2016					
2017					
2018					
2019					
2022					
2023					

Grade boundaries - Raw Score

Year	E	D	C	B	A	Max UMS
2016	16	22	28	35	42	56
2017	14	21	29	37	45	61
2018	19	26	33	41	49	65
2019	19	26	33	40	48	64
2022	14	22	30	38	47	65
2023	13	19	25	32	39	53

Surname	Centre Number	Candidate Number
Other Names		2

GCE AS/A Level

2410U20-1 – **NEW AS**

S16-2410U20-1

CHEMISTRY – Unit 2
Energy, Rate and Chemistry of Carbon Compounds

P.M. FRIDAY, 10 June 2016

1 hour 30 minutes

	For Examiner's use only		
	Question	**Maximum Mark**	**Mark Awarded**
Section A	**1. to 6.**	**10**	
Section B	**7.**	**11**	
	8.	**13**	
	9.	**9**	
	10.	**9**	
	11.	**18**	
	12.	**10**	
	Total	**80**	

2410U201
01

ADDITIONAL MATERIALS

In addition to this examination paper, you will need a:
- calculator;
- **Data Booklet** supplied by WJEC.

INSTRUCTIONS TO CANDIDATES

Use black ink or black ball-point pen. Do not use gel pen or correction fluid.

Write your name, centre number and candidate number in the spaces at the top of this page.

Section A Answer **all** questions in the spaces provided.
Section B Answer **all** questions in the spaces provided.

Candidates are advised to allocate their time appropriately between **Section A (10 marks)** and **Section B (70 marks)**.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets at the end of each question or part-question.

The maximum mark for this paper is 80.

Your answers must be relevant and must make full use of the information given to be awarded full marks for a question.

The assessment of the quality of extended response (QER) will take place in **Q.8**(*b*).

If you run out of space, use the additional page(s) at the back of the booklet, taking care to number the question(s) correctly.

JUN162410U20101

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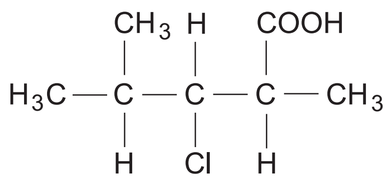
SECTION A

Answer **all** *questions in the spaces provided.*

1. *(a)* Name the compound whose formula is shown below. [1]



....

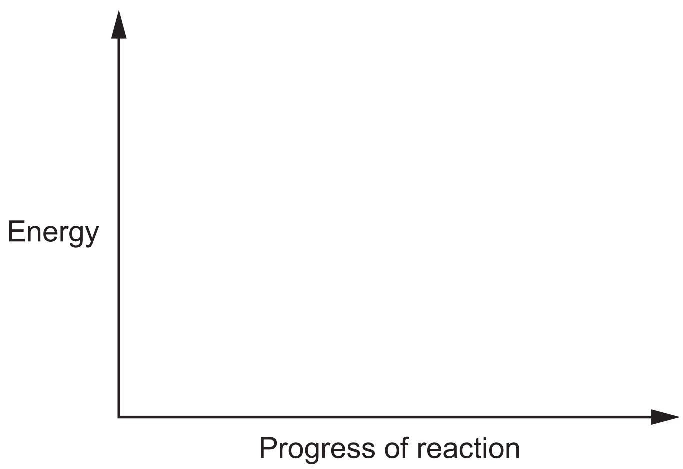
(b) Draw the skeletal formula of the compound shown in *(a)*. [1]

2. Respiration involves the release of energy by using foods. The equation for the respiration of glucose is shown below.

$$C_6H_{12}O_6 \ + \ 6O_2 \longrightarrow 6CO_2 \ + \ 6H_2O \qquad \Delta H = -2900 \text{ kJ mol}^{-1}$$

Photosynthesis is the process by which plants store energy in foods. It can be thought of as the reverse of respiration.

On the axes below sketch and label the energy profile for the **photosynthesis** process. You should label the enthalpy change, ΔH, and the activation energy, E_a. [2]



Examiner only

3. Draw a section of the addition polymer formed from the monomer below.

$$CH_3CH(OH)CH{=}CHCl$$

You should show **two** repeat units. [1]

4. Bromine reacts with methanoic acid according to the equation below.

$$Br_2(aq) + HCOOH(aq) \longrightarrow 2Br^-(aq) + 2H^+(aq) + CO_2(g)$$

A student wanted to follow the rate of the reaction and mixed solutions of known concentration of bromine and methanoic acid.

Suggest **two** methods which the student could use to follow the rate of this reaction as it proceeds. [2]

Method 1 ....

....

Method 2 ....

....

2410U201 03

Examiner only

5. *(a)* State what feature of a molecule gives rise to *E-Z* isomerism. [1]

....

(b) Does the molecule shown below show *E-Z* isomerism? Explain your answer. [1]

$$CH_3CH_2CH{=}C(CH_3)_2$$

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6. On warming, copper(II) oxide reacts with aqueous methanoic acid. Write the equation for this reaction. [1]

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10

Examiner only

SECTION B

Answer **all** *questions in the spaces provided.*

7. Sian was given the following instructions for an experiment to find the enthalpy change for the reaction below.

$$Mg(s) + 2HCl(aq) \longrightarrow MgCl_2(aq) + H_2(g)$$

Instructions

- Weigh the magnesium.
- Place $25.0\,cm^3$ of $0.5\ mol\ dm^{-3}$ hydrochloric acid in an insulating plastic container.
- Measure the initial temperature of the acid in the container.
- Add the magnesium to the acid and stir.
- Measure the highest temperature reached.

(a) Draw a labelled diagram showing how the apparatus is set up to carry out this experiment. You should show how the heat losses could be minimised. [3]

Examiner only

(b) Sian recorded the following data.

Mass of magnesium strip = 0.1 g

Initial temperature of acid = 21.0 °C

Highest temperature reached = 35.5 °C

Use these data to calculate the enthalpy change for the reaction, Δ*H*.

$$Mg(s) + 2HCl(aq) \longrightarrow MgCl_2(aq) + H_2(g)$$

Assume that 4.18 J is needed to raise the temperature of 1.0 cm^3 of all aqueous solutions by 1.0 °C and that the hydrochloric acid was in excess.

Give your answer in kJ mol^{-1}. [3]

2410U201 07

Δ*H* = ... kJ mol^{-1}

(c) Efan repeated the experiment using 0.20 g of magnesium and 50.0 cm^3 of 0.1 mol dm^{-3} hydrochloric acid. Although he used his data correctly to calculate the enthalpy change of reaction for 1 mol of magnesium, his answer was numerically much less than that obtained by Sian. By using the data above and that in part *(b)*, explain why Efan obtained a different value for Δ*H*. [2]

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Examiner only

(d) Sian used a one-decimal place balance to weigh the magnesium. Calculate the maximum percentage error in the mass of magnesium she used. Show clearly how you obtained your answer. [2]

Maximum percentage error = ... %

(e) Burning magnesium, when placed in a gas-jar containing chlorine, forms magnesium chloride and energy is released.

$$Mg(s) + Cl_2(g) \longrightarrow MgCl_2(s)$$

Suggest why this reaction cannot be used to measure the enthalpy change for the reaction that forms magnesium chloride from magnesium. [1]

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11

Examiner only

8. *(a)* (i) Define the term *standard enthalpy change of combustion*, $\Delta_c H^\theta$. [2]

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(ii) The standard enthalpy change of combustion of ethane is –1561 kJ mol^{-1}.

The values of some average bond enthalpies are shown in the table below.

Bond	Average bond enthalpy / kJ mol^{-1}
C—C	348
O═O	495
C═O	799
O—H	463

$$C_2H_6 \ + \ \ldots\ldots\ldots \ O_2 \longrightarrow 2CO_2 \ + \ 3H_2O$$

Balance the equation for the combustion of ethane and use the information given to calculate the average bond enthalpy for a C—H bond. [4]

Bond enthalpy = ... kJ mol^{-1}

(iii) Suggest a reason why bond enthalpies are described as being **average** bond enthalpies. [1]

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2410U201 09

Examiner only

(b) Charcoal consists mainly of carbon. It has been produced for many centuries by heating wood in the absence of oxygen. Natural gas consists mainly of methane and is obtained from underground sources. It was formed in a similar way to coal and oil.

The enthalpy changes of combustion of carbon and methane are in the table.

Substance	Enthalpy change of combustion / kJ mol^{-1}
carbon, C	–394
methane, CH_4	–889

Two students were discussing the use of charcoal and methane as fuels.

One said that 'methane produced more heat per gram when burned so that it was a better fuel'.

The other student said that 'the use of charcoal contributed less to the overall increase of carbon dioxide levels in the atmosphere'.

Discuss these two statements. [6 QER]

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Examiner only

9. All hydrocarbons can be burned but, apart from in combustion reactions, alkenes are more reactive than alkanes.

(a) Describe the bonding in propene and use this to explain its reactivity. [5]

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(b) Draw the mechanism for the reaction between propene and bromine. You should show any relevant dipoles, lone pairs of electrons and curly arrows to indicate the movement of pairs of electrons. [4]

9

Examiner only

10. Halogenoalkanes are hydrolysed when heated with aqueous sodium hydroxide.

(a) (i) Write the equation to show the hydrolysis of 3-chloro-2-methylpentane with aqueous sodium hydroxide. You should show clearly the structure of the organic reagent and product. [1]

(ii) State the name of the mechanism for the reaction taking place in part (i). [1]

....

(iii) Describe a chemical test to show that this hydrolysis reaction has occurred. Include the test and the result expected. [2]

....

....

Examiner only

(b) The rate of reaction between a halogenoalkane and aqueous sodium hydroxide was measured when different concentrations of reagents were used. The results are shown in the table.

Experiment	Initial concentration $OH^-(aq)$ / mol dm^{-3}	Initial concentration halogenoalkane / mol dm^{-3}	Initial rate of reaction / mol dm^{-3} s^{-1}
1	0.10	0.10	1.20×10^{-5}
2	0.10	0.20	2.40×10^{-5}
3	0.10	0.30	3.60×10^{-5}
4	0.20	0.10	1.20×10^{-5}
5	0.30	0.10	1.20×10^{-5}

(i) Use these data to deduce how the concentration of each reagent affects the rate of reaction. Explain how you reached your conclusions. [2]

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(ii) What would be the effect on the initial rate of reaction if the halogen in the halogenoalkane were changed from chlorine to bromine? You should assume that this is the only change made. Explain your answer. [3]

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Examiner only

11. *(a)* Esters are formed when carboxylic acids are heated with alcohols in the presence of concentrated sulfuric acid. This is a reversible reaction that is carried out by heating the reagents under reflux.

(i) Draw a labelled diagram of the apparatus you would use to carry out a reaction under reflux. [3]

(ii) Explain how the apparatus that you have drawn in part (i) results in the process of reflux. [1]

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(iii) Why is it necessary to use a reflux technique in this type of reaction? [1]

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(iv) State the function of the concentrated sulfuric acid in this reaction. [1]

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Examiner only

(b) The equation shows the reaction between ethanoic acid and methanol.

$$CH_3COOH + CH_3OH \rightleftharpoons CH_3COOCH_3 + H_2O$$

When 3.00 g of ethanoic acid was heated under reflux with 1.28 g of methanol in the presence of concentrated sulfuric acid for 10 minutes it was found that 1.18 g of methyl ethanoate was formed.

(i) Suggest how the methyl ethanoate could be separated from the mixture formed in the reaction vessel. [1]

....

(ii) Calculate the percentage of theoretical yield of methyl ethanoate obtained. Show your working. [3]

Percentage yield = ... %

(iii) Suggest **one** change that could be made to this preparation to improve the percentage yield of methyl ethanoate. Explain your suggestion. [2]

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Examiner only

(c) Two reactions of organic compound **A** are shown below.

$$\mathbf{B} \xleftarrow{\text{conc } H_2SO_4} \mathbf{A} \xrightarrow{Cr_2O_7^{2-}/H^+} \mathbf{C}$$

Compound **B** is a straight-chain hydrocarbon with the formula C_4H_8.

(i) What type of reaction is taking place when compound **A** is heated with concentrated sulfuric acid to form compound **B**? [1]

....

(ii) Draw the displayed formulae of **two** possible isomers of **A**. [2]

(iii) What colour **change** is seen when compound **A** reacts with the $Cr_2O_7^{2-}/H^+$ solution? [1]

....

(iv) What type of reaction is taking place when compound **A** is heated with $Cr_2O_7^{2-}/H^+$ to form compound **C**? [1]

....

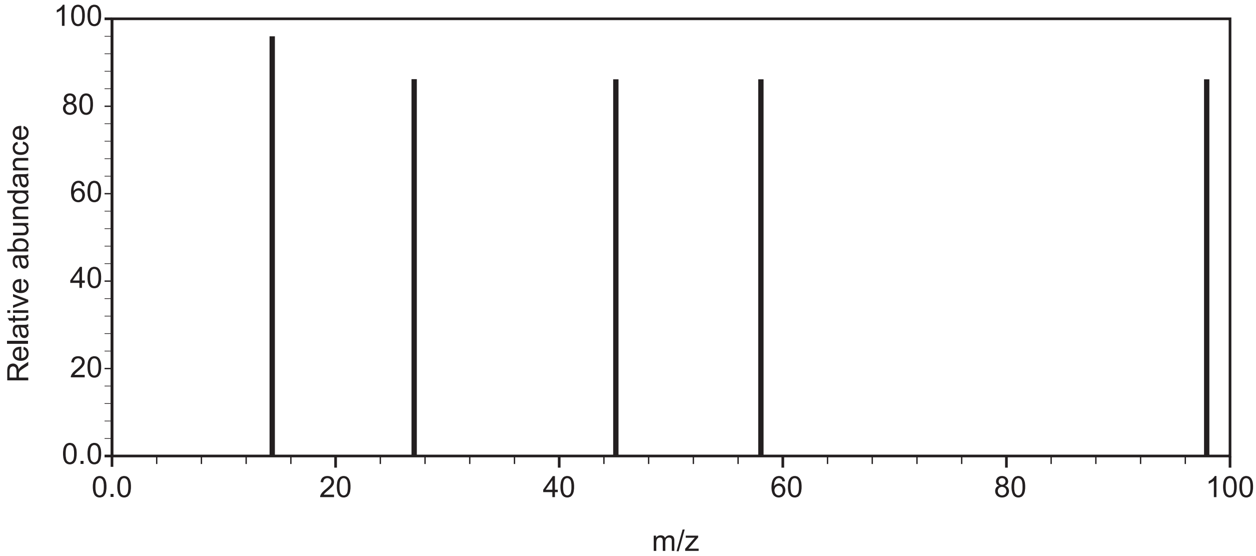
(v) Draw a displayed formula of compound **C**. [1]

18

Examiner only

12. Chemists are investigating an unknown compound **X**. They obtain information from a variety of sources.

Compound **X** contains 61.2 % carbon, 6.1 % hydrogen and 32.7 % oxygen by mass. A simplified mass spectrum of compound **X** is shown below.



When solid sodium carbonate is added to an aqueous solution of **X** effervescence is observed.

There are 5 peaks in the ^{13}C NMR spectrum of **X**.

1 mol of **X** reacts completely with 320 g of bromine in the dark.

Use **all** the data given to find the structure of compound **X**. Explain what information can be found from each piece of data. [10]

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10

END OF PAPER

Surname	Centre Number	Candidate Number
Other Names		2

GCE AS/A LEVEL – NEW

2410U20-1

S17-2410U20-1

CHEMISTRY – AS unit 2
Energy, Rate and Chemistry of Carbon Compounds

FRIDAY, 9 JUNE 2017 – AFTERNOON

1 hour 30 minutes

	For Examiner's use only		
	Question	**Maximum Mark**	**Mark Awarded**
Section A	**1. to 6.**	**10**	
Section B	**7.**	**14**	
	8.	**16**	
	9.	**13**	
	10.	**14**	
	11.	**13**	
	Total	**80**	

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JUN172410U20101

Examiner only

SECTION A

Answer **all** *questions in the spaces provided.*

1. *(a)* Draw the **displayed** formula of *(E)*-pent-2-ene. [1]

(b) Draw the **skeletal** formula of 3-chlorohex-1-ene. [1]

2. State the molecular formula of a straight-chained saturated halogenoalkane whose molecules each contain 50 carbon atoms and 2 chlorine atoms. [1]

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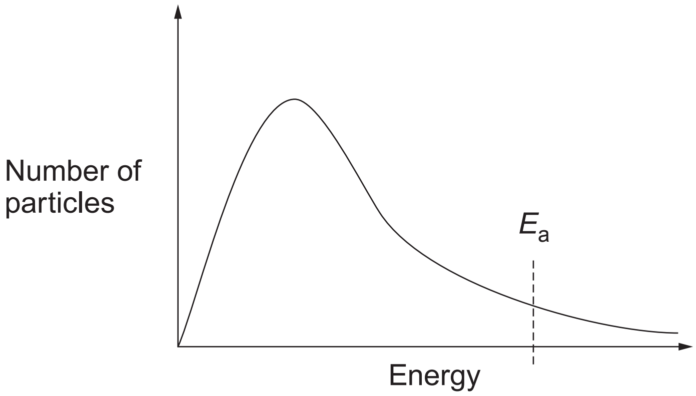
Examiner only

3. The diagram shows the distribution of the energies of particles in a gas at a particular temperature.



On the diagram draw a curve to show the distribution of the energies of the same particles at a higher temperature.

Use the diagram to explain the effect of increasing temperature on the rate of a reaction. [2]

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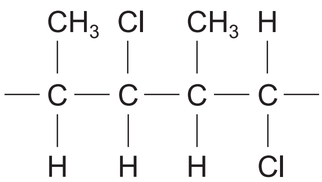
4. A section of polymer is shown.



Draw the structural formula of the monomer used to make this polymer. [1]

Examiner only

5. Complete the statement below. [2]

To test for the presence of iodine in an organic compound the compound is first heated with .... This is followed by adding .... and then .... to the mixture. If iodine was present in the organic compound then a .... precipitate is seen.

6. *(a)* Complete the equation below to show clearly the missing product. [1]

(b) Name the catalyst used in the reaction in part *(a)*. [1]

....

10

Examiner only

SECTION B

*Answer **all** questions in the spaces provided.*

7. *(a)* Ethanol, C_2H_5OH, is a liquid at room temperature. Write the equation that corresponds to the standard enthalpy change of formation of ethanol. [2]

....

(b) Why is it difficult to measure the standard enthalpy change of formation of ethanol directly? [1]

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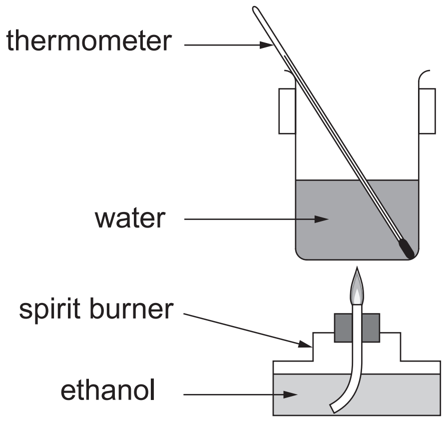
....

(c) The following apparatus can be used to determine the enthalpy change of combustion of ethanol, $\Delta_c H$.



(i) Explain how the enthalpy change of combustion is calculated from the results collected using this apparatus. [3]

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2410U201
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(ii) Why is it difficult to use this apparatus to determine the enthalpy change of combustion of ethane, C_2H_6? [1]

....

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(d) Two students carried out an experiment using the apparatus in part *(c)*.

Neither followed the given method correctly. Euan extinguished the flame after only 10 seconds. Carys continued to heat for two minutes after the water had boiled.

Explain the effect that each of the students' errors had on the values they obtained in this experiment. [2]

Effect of Euan's error

....

....

Effect of Carys' error

....

....

Examiner only

(e) Standard enthalpy changes of formation, $\Delta_f H^\theta$, can be found by using standard enthalpy changes of combustion, $\Delta_c H^\theta$.

The table shows some standard enthalpy changes of combustion.

Substance	Standard enthalpy change of combustion / kJ mol^{-1}
carbon, C	–394
hydrogen, H_2	–286
ethanol, C_2H_5OH	–1371

(i) Use these data to calculate the standard enthalpy change of formation of ethanol. Show clearly how you carried out your calculation. [3]

$\Delta_f H^\theta$ = ... kJ mol^{-1}

(ii) Amir said that hydrogen is the best fuel of these three in terms of the energy released by burning equal masses. Is he correct? Justify your answer. [2]

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14

2410U201 07

Examiner only

8. *(a)* Compound **X** is a hydrocarbon that contains 85.6% of carbon by mass. The relative molecular mass of **X** is in the range 50-60.

11.2 g of compound **X** decolourised exactly 32.0 g of bromine in the absence of light. Further bromine was decolourised when the reaction mixture was placed in direct sunlight.

(i) Find the empirical formula of **X** and hence its molecular formula. Show clearly how you carried out your calculation. [3]

Molecular formula ....

(ii) Explain what can be deduced from the fact that 11.2 g of **X** reacts with exactly 32.0 g of bromine. [2]

....

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(iii) What type of reaction mechanism occurs when **X** reacts with bromine in the absence of light? [1]

....

(iv) What type of reaction mechanism occurs when **X** reacts with more bromine in the presence of sunlight? [1]

....

Examiner only

(v) Draw displayed formulae for each of the following. [3]

Compound **X**

The product formed when **X** reacts with bromine in the absence of light

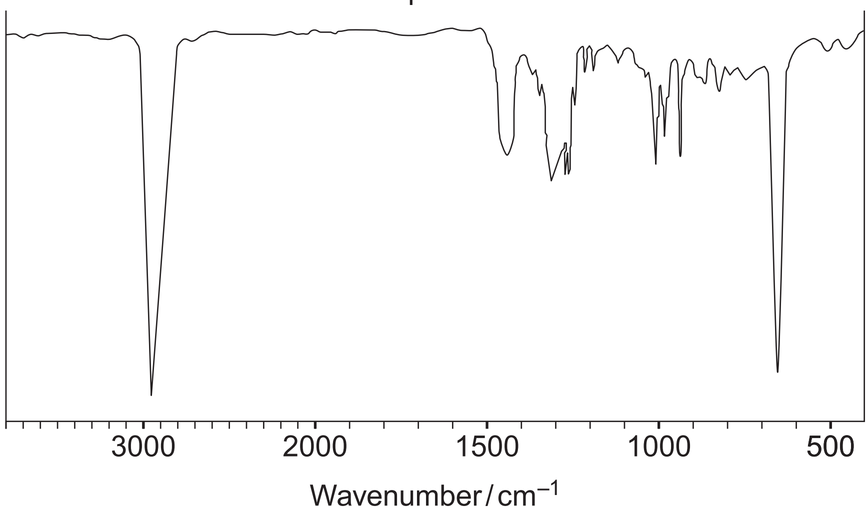
One possible product formed when **X** reacts with excess bromine in the presence of sunlight

2410U201 09

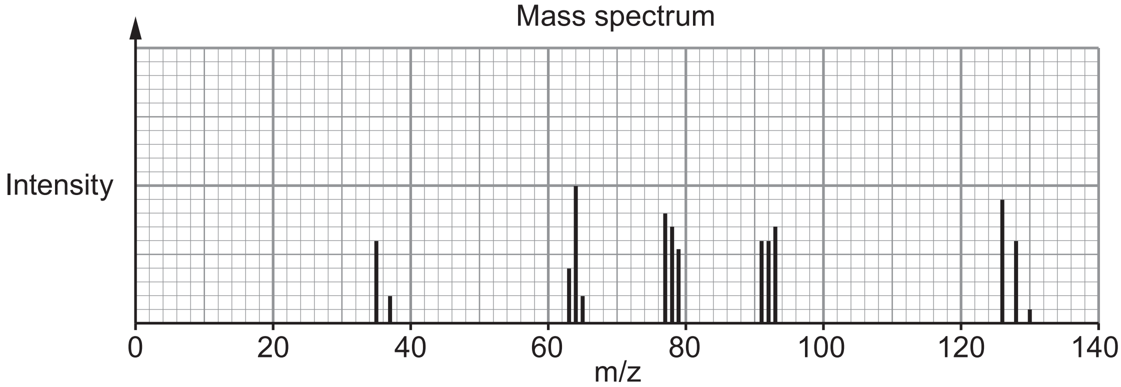
Examiner only

(b) Compound **Y** is a halogenoalkane. A simplified form of the IR spectrum, the mass spectrum, the ^{13}C NMR spectrum and the low resolution ^{1}H NMR spectrum of **Y** are shown below.

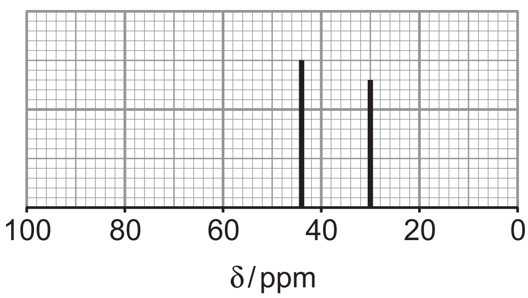
IR spectrum



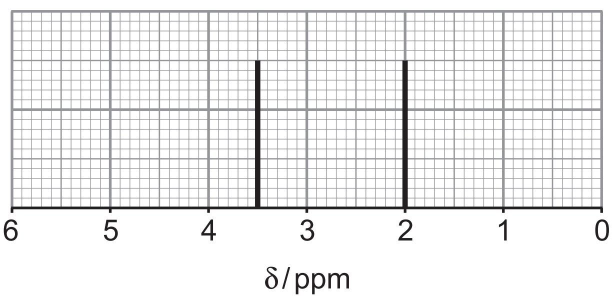
Mass spectrum



^{13}C NMR



^{1}H NMR



Examiner only

Use these spectra to determine the identity of compound **Y**. You must use evidence from each spectrum and explain how you used it. [6 QER]

16

Examiner only

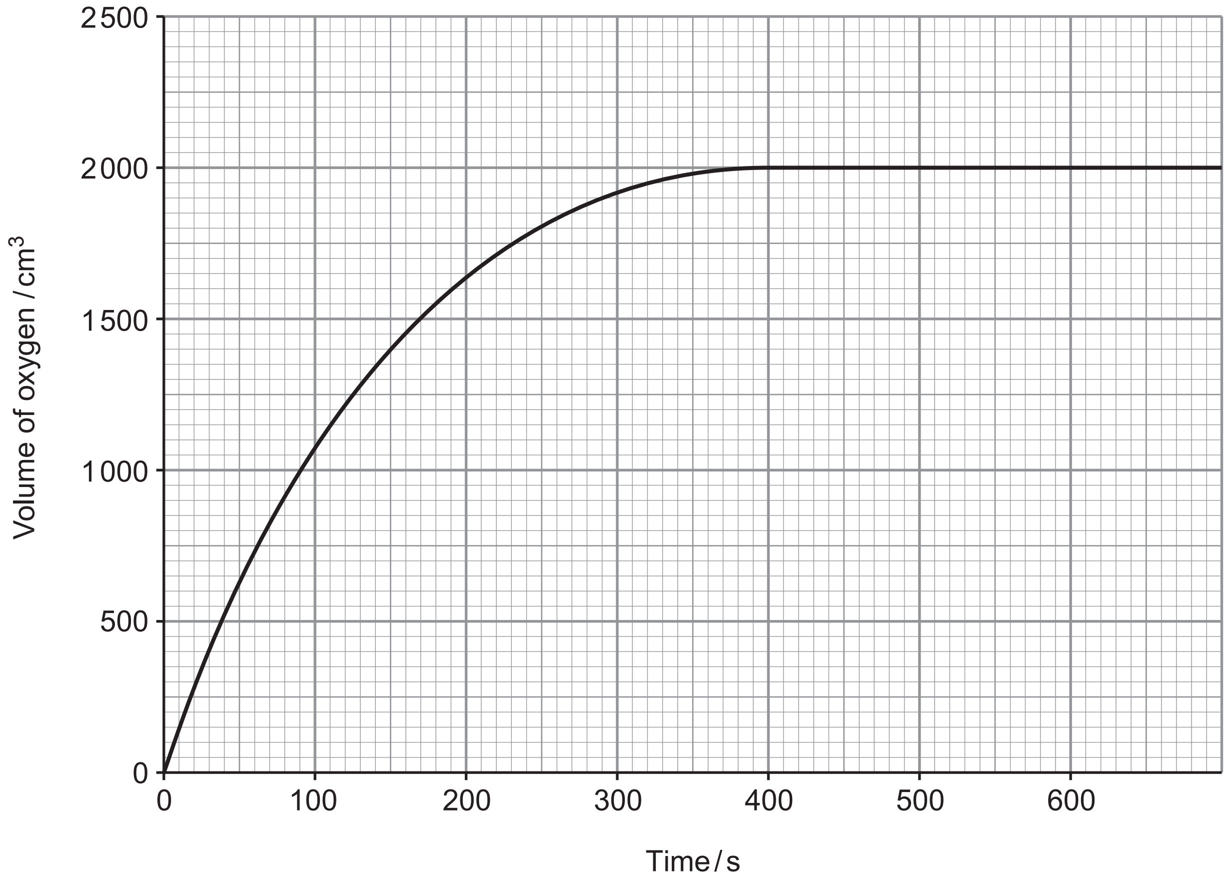
9. A student was investigating the rate of reaction for the catalysed decomposition of hydrogen peroxide to form oxygen.

$$2H_2O_2(aq) \longrightarrow O_2(g) + 2H_2O(l)$$

The student decided to follow the reaction by measuring the volume of oxygen produced at fixed time intervals. He added a spatula measure of manganese(IV) oxide to 100 cm^3 of aqueous hydrogen peroxide solution and started the timer.

(a) Draw a labelled diagram of the apparatus he could use to carry out the experiment and measure the volume of oxygen produced. [2]

(b) The student plotted his results to produce a graph as shown.



Examiner only

(i) Use the graph to calculate the initial rate of the reaction. Show clearly, on your graph, how you obtained your answer. [2]

Initial rate = ... $cm^3\ s^{-1}$

(ii) Use the graph to calculate the rate of the reaction at 200 seconds. Show clearly, on your graph, how you obtained your answer. [1]

Rate = ... $cm^3\ s^{-1}$

(iii) Use collision theory to explain why the rates you have calculated in parts (i) and (ii) are different. [2]

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(iv) Use the graph to calculate the concentration, in mol dm^{-3}, of the hydrogen peroxide solution used by the student. Give your answer correct to **three** significant figures. [3]

Assume that the reaction was carried out at 298 K and 1 atm pressure.

Concentration = ... mol dm^{-3}

Examiner only

(c) A second student said that she had seen cobalt(II) chloride speed up the rate of production of oxygen very significantly in a different reaction.

Describe how the students could determine if cobalt(II) chloride is a better catalyst than manganese(IV) oxide in the decomposition of hydrogen peroxide. [3]

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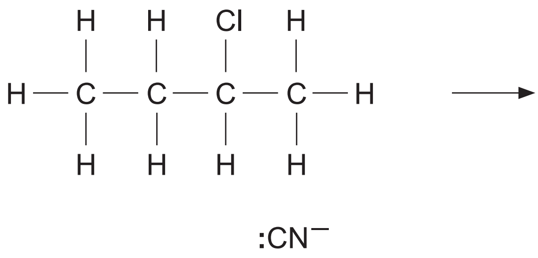
13

Examiner only

10. *(a)* Halogenoalkanes react with potassium cyanide in a similar way to their reaction with OH^-(aq).

(i) Complete the mechanism of the reaction between 2-chlorobutane and potassium cyanide. You should include relevant dipoles and curly arrows to show the movement of electrons. [4]



(ii) Name the type of reaction mechanism shown in part (i). [1]

....

(b) When 2-chlorobutane is heated with sodium hydroxide dissolved in ethanol three different organic products are formed.

(i) Draw the displayed formulae of the **three** products. Name each product. [3]

(ii) Name the type of reaction taking place. [1]

....

Examiner only

(c) Until recently chlorofluorocarbons, CFCs, were used in many commercial products but nowadays hydrofluorocarbons, HFCs, are preferred.

Explain why HFCs are preferred to CFCs. You do **not** need to include the mechanism or equation for any reaction which you describe. [5]

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14

Examiner only

11. *(a)* Ethanol can be prepared by the fermentation of glucose, $C_6H_{12}O_6$. The equation for this reaction is as follows.

$$C_6H_{12}O_6(aq) \longrightarrow 2C_2H_5OH(aq) + 2CO_2(g)$$

(i) A student wanted to prepare ethanol and was given the following instructions.

- Mix glucose and water.
- Put the mixture into a stoppered bottle.
- Leave the bottle for several weeks.
- Separate off the ethanol.

Suggest how the student should modify these instructions in order to obtain a good yield of pure ethanol. [3]

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(ii) Calculate the atom economy of this process as a method of preparing ethanol. [2]

Atom economy = ... %

(iii) Ethanol can also be made by the hydration of ethene.

Describe **one** environmental consideration in choosing between fermentation and hydration as methods for ethanol production. [2]

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Examiner only

(b) Alcohols are classified as being primary, secondary or tertiary.

(i) Explain the meaning of *primary* as applied to alcohols. [1]

....

....

(ii) Give the structural formula and name of a tertiary alcohol that contains five carbon atoms. [2]

Name ....

(iii) Describe **one** test that can be used to investigate whether an alcohol is secondary or tertiary. Include the reagents used and the observations for both types of alcohol. [3]

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13

END OF PAPER

Surname	Centre Number	Candidate Number
Other Names		2

GCE AS/A LEVEL

2410U20-1

S18-2410U20-1

CHEMISTRY – AS unit 2

Energy, Rate and Chemistry of Carbon Compounds

FRIDAY, 25 MAY 2018 – MORNING

1 hour 30 minutes

	For Examiner's use only		
	Question	**Maximum Mark**	**Mark Awarded**
Section A	**1. to 7.**	**10**	
Section B	**8.**	**11**	
	9.	**12**	
	10.	**16**	
	11.	**12**	
	12.	**13**	
	13.	**6**	
	Total	**80**	

2410U201
01

ADDITIONAL MATERIALS

In addition to this examination paper, you will need a:
- calculator;
- **Data Booklet** supplied by WJEC.

INSTRUCTIONS TO CANDIDATES

Use black ink or black ball-point pen. Do not use gel pen or correction fluid.

Write your name, centre number and candidate number in the spaces at the top of this page.

Section A Answer **all** questions in the spaces provided.
Section B Answer **all** questions in the spaces provided.

Candidates are advised to allocate their time appropriately between **Section A (10 marks)** and **Section B (70 marks)**.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets at the end of each question or part-question.

The maximum mark for this paper is 80.

Your answers must be relevant and must make full use of the information given to be awarded full marks for a question.

The assessment of the quality of extended response (QER) will take place in **Q.10**_(a)_(i).

If you run out of space, use the additional page(s) at the back of the booklet, taking care to number the question(s) correctly.

Examiner only

SECTION A

*Answer **all** questions in the spaces provided.*

1. Draw the structure of a secondary alcohol that contains 6 carbon atoms. [1]

2. Draw the **skeletal** formula of 3-chloro-2,2-dimethylpentane. [1]

3. Explain why ethanol is soluble in water but ethane is not. [2]

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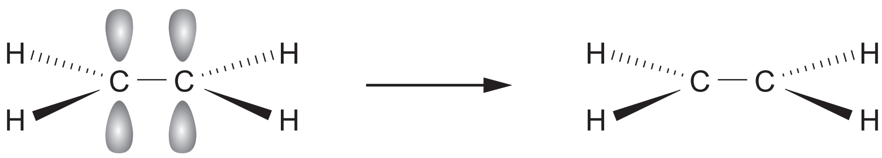
4. Complete the diagram to show the formation of the π-bond in ethene. [1]



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5. *(a)* Complete the equation for the fermentation of glucose. [1]

$C_6H_{12}O_6 \longrightarrow$ .... + ...

(b) Name the substance generally used to catalyse this reaction. [1]

....

6. Draw **two** repeat units for the polymer formed from the monomer pent-2-ene. [1]

7. Draw diagrams to show the structures of the *E* and *Z* isomers of 2-bromopent-2-ene.
Label the isomers *E* and *Z*. [2]

2410U201
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SECTION B

*Answer **all** questions in the spaces provided.*

8. A student was told that he could prepare chloroethane, C_2H_5Cl, by mixing ethane with chlorine. He added 2.0 g of ethane to excess chlorine and left the mixture exposed to ultraviolet light for several hours. He was then able to use a university laboratory to see whether chloroethane had been made.

(a) State an instrumental method by which the sample could be analysed. Explain how this would show that chlorination had occurred. [2]

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(b) State why it is necessary to use ultraviolet light when making chloroethane from ethane. Give equations to show the mechanism for the formation of chloroethane. [4]

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Examiner only

(c) The student found that he had made 1.0 g of chloroethane. Calculate his percentage yield. [3]

Percentage yield = ... %

(d) The student was disappointed by the yield he obtained but his teacher told him that the yield was always poor due to other products being formed in this reaction.

Identify **two** other organic products, apart from chloroethane, and explain how they are formed. [2]

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9. *(a)* (i) The average bond enthalpy of a C—C bond is quoted as 348 kJ mol^{-1}.

Explain what is meant by *bond enthalpy*. [2]

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(ii) Ethyne, C_2H_2, contains a C≡C. It reacts with hydrogen in a similar way to ethene.

$$H{-}C{\equiv}C{-}H \quad + \quad 2H_2 \longrightarrow C_2H_6$$

Some average bond enthalpies are given in the table.

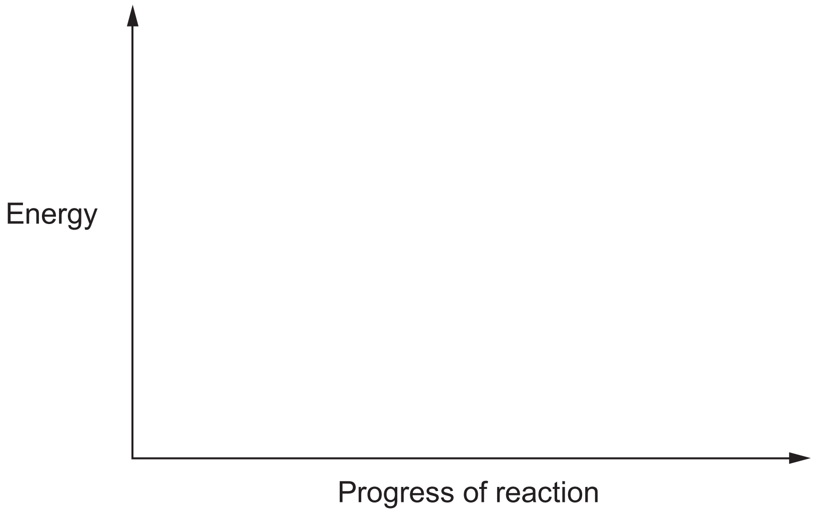
Bond	Average bond enthalpy / kJ mol^{-1}
C≡C	839
C—C	348
C—H	413
H—H	436

Use the data to calculate the enthalpy change, Δ*H*, for the reaction of ethyne and hydrogen. [3]

Δ*H* = ... kJ mol^{-1}

Examiner only

(iii) Use your answer to part (ii) to sketch an energy profile diagram for this reaction on the axes below. Label ΔH and the activation energy, E_a, on your diagram. [2]



(b) Enthalpy changes of reaction are often found indirectly. The enthalpy change for the reaction of ethyne with hydrogen, as shown in part *(a)*, can be determined by using enthalpy changes of combustion.

The table gives some enthalpy changes of combustion, $\Delta_c H^\theta$.

Substance	Enthalpy change of combustion, $\Delta_c H^\theta$ / kJ mol^{-1}
hydrogen, H_2	–286
ethyne, C_2H_2	–1300
ethane, C_2H_6	–1600

Use these enthalpy changes to calculate the enthalpy change, ΔH, for the reaction of ethyne and hydrogen. [3]

$$C_2H_2 \quad + \quad 2H_2 \longrightarrow C_2H_6$$

ΔH = ... kJ mol^{-1}

2410U201
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(c) The theoretical values that you have calculated in parts *(a)*(ii) and *(b)* are both the enthalpy change for the reaction between ethyne and hydrogen.

Suggest a reason why these values are not the same. [1]

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(d) Suggest the type of reaction that occurs between ethyne and hydrogen. [1]

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12

10. *(a)* Halogenoalkanes can be hydrolysed using water in a similar way to using aqueous sodium hydroxide.

A student carried out an experiment to investigate the rate of reaction for the hydrolysis of halogenoalkanes using water. The student used aqueous ethanol to dissolve the halogenoalkane and then added a few drops of aqueous silver nitrate. He timed how long it took to produce a precipitate. He obtained the results shown in the table.

Halogenoalkane	Time / s
1-chloropropane, C_3H_7Cl	300
1-bromopropane, C_3H_7Br	90
1-iodopropane, C_3H_7I	15

The student tried to explain these results and he looked on the internet to find the following data.

Bond	Bond enthalpy / $kJ\,mol^{-1}$
C—H	413
C—C	348
C—F	485
C—Cl	328
C—Br	276
C—I	240

Element	Electronegativity
chlorine	3.16
bromine	2.96
iodine	2.66
carbon	2.55

Examiner only

(i) Use both sets of data to explain the student's results. Include an equation to show the hydrolysis reaction between a halogenoalkane and aqueous sodium hydroxide and name this type of reaction mechanism. [6 QER]

(ii) Write an **ionic** equation, including state symbols, for a reaction that produces a silver halide precipitate. [1]

(iii) Suggest a practical method by which the student could have obtained these results. [2]

(iv) Suggest the difference that the student would have observed in his experiments if he had not added ethanol to the water before adding the halogenoalkane. [1]

2410U201
11

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(b) Chlorofluorocarbons, CFCs, were historically used for a variety of commercial and domestic purposes but nowadays their use is very restricted.

(i) Outline the adverse environmental effects of the use of CFCs.

You do not need to include any equations for the reactions involved. [4]

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(ii) Use relevant data in part *(a)* to explain why hydrofluorocarbons, HFCs, have replaced CFCs in many of their uses. [2]

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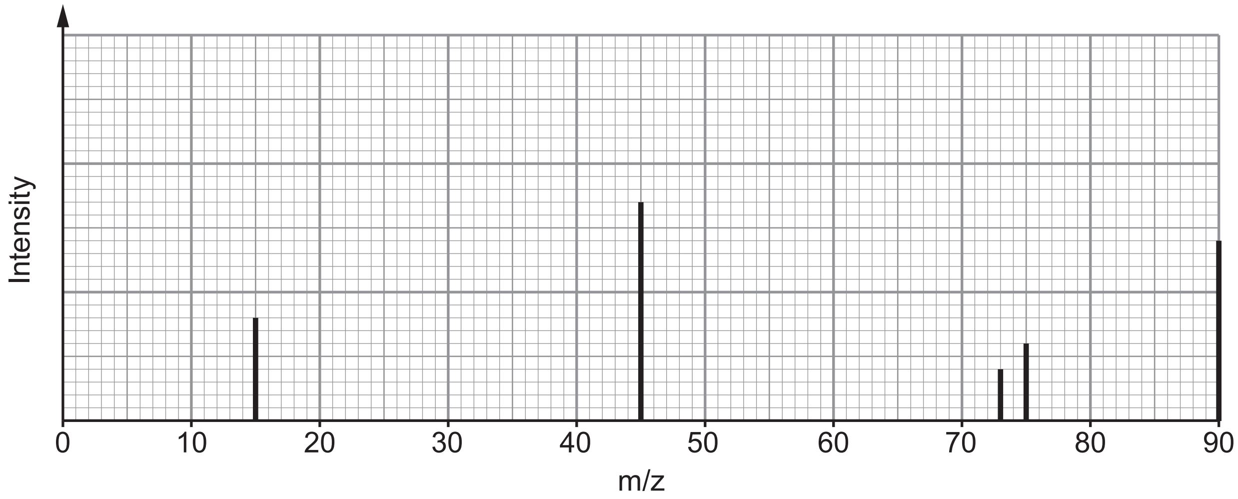
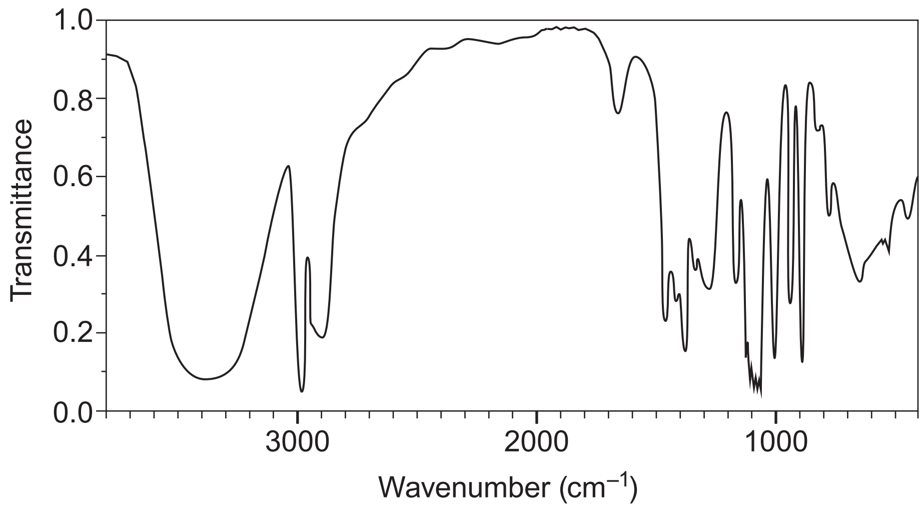
16

11. Compound **X** contains only carbon, hydrogen and oxygen. On analysis it was found to contain 53.3% carbon and 11.1% hydrogen by mass.

A simplified form of the mass spectrum and the infrared absorption spectrum for **X** are shown.





The low resolution ^{1}H NMR spectrum of **X** has three peaks.

When **X** is warmed with excess acidified potassium dichromate(VI) there is a colour change. The organic product of this reaction does **not** react with aqueous sodium carbonate.

Examiner only

(a) Use **all** the data given to find the structure of compound **X**. Explain what information can be found from each piece of data. [10]

Structure of **X**

Examiner only

(b) (i) State the type of reaction that occurs when **X** is warmed with acidified potassium dichromate(VI). [1]

....

(ii) Draw the structure of the organic product formed when **X** reacts with acidified potassium dichromate(VI). [1]

12

Examiner only

12. *(a)* Iodide ions can be oxidised to iodine by reaction with acidified hydrogen peroxide.

$$H_2O_2 \quad + \quad 2I^- \quad + \quad 2H^+ \longrightarrow I_2 \quad + \quad 2H_2O$$

The rate of reaction can be followed in a clock reaction by the appearance of a blue-black colour.

An experiment was carried out to determine the effect on the rate of reaction of varying the concentration of iodide ions. All other volumes and concentrations were kept constant. The results are shown in the table.

Concentration I^- / mol dm^{-3}	Time for appearance of blue-black colour / s	Rate / $s^{-1} \times 1000$
0.1	56	18
0.2	20	....
0.3	18	....
0.4	12	....
0.5	10	....

(i) Use rate = $\dfrac{1000}{\text{time}}$ to calculate the rate for each experiment and complete the table. [1]

Examiner only

(ii) On the axes below, plot the concentration of I^- against rate and draw a suitable line. [3]

(iii) From the graph deduce the relationship between concentration of I^- and rate of reaction. [1]

....

....

(iv) **Use the graph** to calculate the time the reaction would take to turn blue-black using a 0.15 mol dm^{-3} solution of I^-. Show clearly how you obtained your answer. [2]

Time = ... s

Examiner only

(v) For each experiment the rate was calculated using the time taken to produce excess iodine. Explain why this is only an approximation for the rate **as the reaction proceeds**. [2]

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(b) (i) Draw a Boltzmann energy distribution curve. Label the axes. [2]

(ii) Use this energy distribution curve to explain how catalysts affect the rate of a reaction. [2]

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Examiner only

13. *(a)* Explain what is meant by a carbon-neutral fuel. [3]

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(b) Ethane and an unknown alkane were burned in oxygen.

It follows from the equation below that one volume of ethane produced two volumes of carbon dioxide and three volumes of water vapour.

$$C_2H_6(g) \quad + \quad \tfrac{7}{2}O_2(g) \quad \longrightarrow \quad 2CO_2(g) \quad + \quad 3H_2O(g)$$

$10\,cm^3$ of the unknown alkane C_xH_y burned according to the following equation.

$$C_xH_y(g) \quad + \quad (x \;+\; \tfrac{y}{4})O_2(g) \quad \longrightarrow \quad xCO_2(g) \quad + \quad \tfrac{y}{2}\,H_2O(g)$$

The total volume of carbon dioxide and water vapour produced was $20\,cm^3$ more than the original volume of C_xH_y and oxygen. All volumes were measured at the same temperature and pressure.

(i) State the volumes of carbon dioxide and water vapour produced on burning $10\,cm^3$ of the unknown alkane in terms of x and y. [1]

Volume carbon dioxide = ... cm^3

Volume water vapour = ... cm^3

(ii) Calculate the value of y. [2]

y = ...

6

END OF PAPER

Surname	Centre Number	Candidate Number
Other Names		2

GCE AS/A LEVEL

2410U20-1

S19-2410U20-1

CHEMISTRY – AS unit 2
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THURSDAY, 23 MAY 2019 – MORNING

1 hour 30 minutes

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	11.	14	
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SECTION A

Answer **all** *questions in the spaces provided.*

1. Draw the **skeletal** formula of 2-bromo-3-methylbutane. [1]

2. Write the equation that corresponds to the standard molar enthalpy change of formation of magnesium carbonate, $MgCO_3$. [2]

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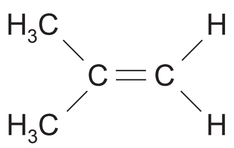
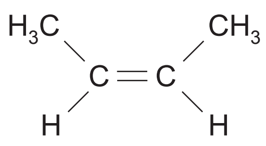
3. The compounds shown below have the same molecular formula.



A



B

State which of these compounds can exist as *E-Z* isomers.

Explain your answer and name the isomer. [2]

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....

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Name of isomer ....

Examiner only

4. Halogenoalkanes are hydrolysed by aqueous sodium hydroxide. State and explain which of 1-fluoropropane, 1-chloropropane and 1-bromopropane is hydrolysed most rapidly. [2]

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5. *(a)* State the type of polymerisation involved when 2-fluorobut-2-ene, $CH_3CF{=}CHCH_3$, forms a polymer. [1]

....

(b) Draw **one** repeat unit of the polymer formed when 2-fluorobut-2-ene is polymerised. [1]

2410U201
03

6. Complete the equation for the formation of ethanol by the fermentation of glucose. [1]

$$C_6H_{12}O_6 \longrightarrow \text{..............................} + \text{..............................}$$

10

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SECTION B

Answer **all** *questions in the spaces provided.*

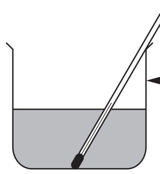
7. *(a)* A student was asked to find the enthalpy change of reaction, ΔH, for the thermal decomposition of calcium hydroxide.

$$Ca(OH)_2(s) \longrightarrow CaO(s) + H_2O(g)$$

This enthalpy cannot be measured directly so Hess's Law is generally used to calculate its value from the enthalpy changes for the reactions of calcium oxide and calcium hydroxide with hydrochloric acid.

The student added a known mass of calcium oxide to hydrochloric acid and measured the temperature change. This was repeated using a known mass of calcium hydroxide. In each experiment $50.0\,cm^3$ of $1.40\,mol\,dm^{-3}$ hydrochloric acid was used.

The diagram shows the apparatus used.



When 1.90 g of calcium oxide was used a temperature rise of 20.5 °C was observed.

(i) The equation for the reaction of calcium oxide with hydrochloric acid is shown.

$$CaO(s) + 2HCl(aq) \longrightarrow CaCl_2(aq) + H_2O(l)$$

Use this equation to show that the hydrochloric acid was in excess in the reaction with calcium oxide. [3]

....

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Examiner only

(ii) Calculate the energy released in this reaction between calcium oxide and hydrochloric acid. [1]

Energy released = ... J

(iii) Calculate the molar enthalpy change for the reaction between calcium oxide and hydrochloric acid. [2]

$$CaO(s) + 2HCl(aq) \longrightarrow CaCl_2(aq) + H_2O(l)$$

Enthalpy change = ... kJ mol^{-1}

sign value

Examiner only

(iv) The value of the molar enthalpy change of reaction for the reaction between calcium hydroxide and hydrochloric acid is –196 kJ mol^{-1}.

Use Hess's Law and your answer to part (iii) to calculate the enthalpy change of reaction for the decomposition reaction.

$$Ca(OH)_2(s) \longrightarrow CaO(s) + H_2O(g)$$

Show clearly how you obtained your answer.

(If you do not have an answer in part (iii), assume that the enthalpy change of reaction is –110 kJ mol^{-1}. This is **not** the correct value.) [2]

Enthalpy change = ... kJ mol^{-1}

(v) Suggest **two** changes to the apparatus shown that would give a more accurate value for the enthalpy changes of the reactions. Give a reason for your answer in both cases. [2]

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Examiner only

(b) Methanol, CH_3OH, is a liquid at room temperature and its molar enthalpy of combustion, $\Delta_c H$, can be found directly.

(i) Write the equation corresponding to the standard enthalpy change of combustion of methanol. [1]

....

(ii) Draw and label suitable apparatus as it is being used in an experiment to determine the enthalpy change of combustion of methanol. [2]

2410U201
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Examiner only

(c) The equation for the combustion of ethene is shown.

$$C_2H_4(g) \ + \ 3O_2(g) \longrightarrow 2CO_2(g) \ + \ 2H_2O(g) \qquad \Delta_cH^\theta = -1387 \text{ kJ mol}^{-1}$$

(i) Use this and the values of the average bond enthalpies in the table to calculate the average bond enthalpy of C—H. [4]

Bond	Average bond enthalpy / kJ mol^{-1}
C═C	614
O═O	495
C═O	799
O—H	465

Average bond enthalpy = ... kJ mol^{-1}

(ii) State why the apparatus you have drawn in part *(b)* cannot be used to determine the enthalpy change of combustion of ethene. [1]

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18

Examiner only

8. Many factors affect the rate of a chemical reaction.

(a) Explain why a change in concentration affects the rate of a reaction. [2]

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(b) Under certain conditions nitrogen monoxide reacts with carbon monoxide and oxygen according to the equation below.

$$NO(g) \ + \ CO(g) \ + \ O_2(g) \longrightarrow NO_2(g) \ + \ CO_2(g)$$

A study of the rate of this reaction, using varying initial concentrations of nitrogen monoxide and oxygen, gave the following data.

Experiment number	Concentration NO / mol dm^{-3}	Concentration O_2 / mol dm^{-3}	Initial rate of reaction / mol $dm^{-3}s^{-1} \times 10^{-4}$
1	1.0×10^{-4}	1.0×10^{-4}	4.40
2	2.0×10^{-4}	1.0×10^{-4}	17.6
3	3.0×10^{-4}	1.0×10^{-4}	39.6
4	2.0×10^{-4}	2.0×10^{-4}	17.6

(i) Use the data to determine how the concentration of NO affects the rate of the reaction. Explain your answer by referring to the experiment numbers. [2]

.....

.....

.....

(ii) Use the data to determine how the concentration of O_2 affects the rate of the reaction. Explain your answer by referring to the experiment numbers. [2]

.....

.....

.....

Examiner only

(iii) Suggest a method that could be used to follow changes over time as the reaction of nitrogen monoxide, carbon monoxide and oxygen proceeds. Explain your answer. [2]

$$NO(g) + CO(g) + O_2(g) \longrightarrow NO_2(g) + CO_2(g)$$

....

....

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(iv) Describe the environmental implications on the atmosphere if the reaction above occurs on a large scale. [2]

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(v) The reaction shown is one of many that are catalysed in a catalytic converter. State where in a car a catalytic converter is used and name a suitable catalyst. [2]

Catalytic converter used in ....

Catalyst used ....

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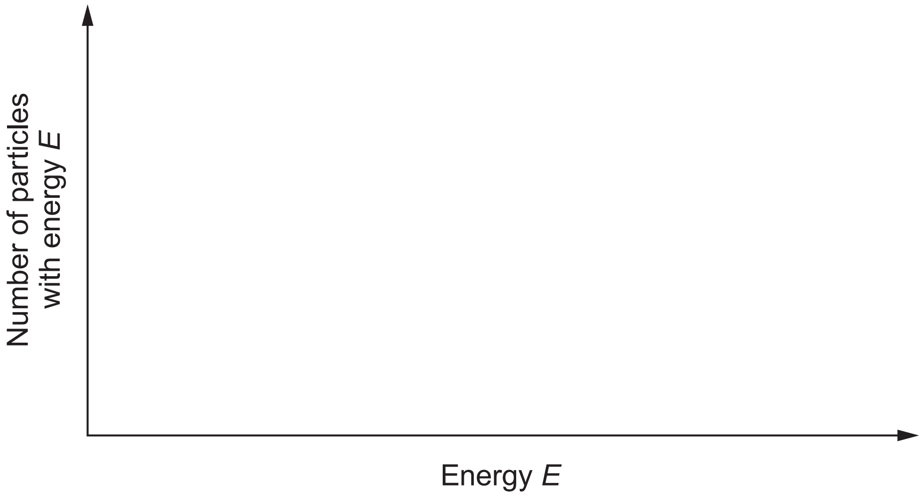
Examiner only

(c) On the axes below draw a Boltzmann distribution curve of the energies of molecules at a certain temperature.

Use this curve to explain how a catalyst increases the rate of a reaction. [3]



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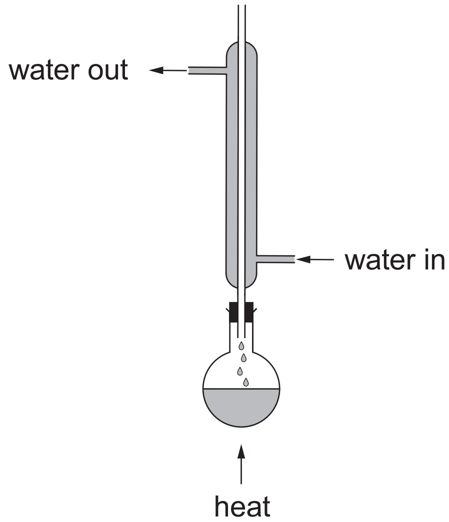
Examiner only

9. *(a)* Alcohols react with carboxylic acids to form esters. To prepare a pure sample of an ester a condenser can be used in the first stage.



(i) Name the method being used with the condenser positioned in this way. Explain why it is necessary to use the condenser. [2]

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(ii) Name the method used to separate the product from the liquid mixture. State which property of an ester allows this method of separation to be used. [2]

Method ....

Property ....

(iii) Name the catalyst most commonly used in the preparation of an ester. [1]

....

(iv) Write the equation for the reaction between methanoic acid and butan-2-ol. Clearly show the structure of the ester formed. [2]

Examiner only

(b) A compound is known to be one of butan-2-ol, $CH_3CH_2CH(OH)CH_3$, 2-methylpropanoic acid, $CH_3CH(CH_3)COOH$, and 3-hydroxybutanoic acid, $CH_3CH(OH)CH_2COOH$.

Choose **two** chemical tests that will enable you to determine which one it is.

Complete the table below.

Give the reagent(s) for each test and the observation expected for a positive result.

For each test put a **tick (✓)** in the box to show the compound that gives a positive result and a **cross (×)** for those that do not. [5]

Reagent(s)	Observation expected for positive result	butan-2-ol	2-methylpropanoic acid	3-hydroxybutanoic acid

12

Examiner only

10. *(a)* Students were discussing the reactivity of organic compounds. One said that reactivity was due to dipoles produced by differences in electronegativity.

Show that this is not correct by discussing the difference in reactivity between alkanes and alkenes. You should include reference to the bonding in both series.

[6 QER]

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Examiner only

(b) In the 19th century Cannizzaro discovered a reaction that involved disproportionation. Such reactions involve the same substance being both oxidised and reduced.

The equation for the Cannizzaro reaction is shown.

$$2C_6H_5CHO + H_2O \longrightarrow C_6H_5CH_2OH + C_6H_5COOH$$

(i) State why this reaction is classified as *disproportionation*. [1]

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....

(ii) A chemist used 9.50 g of C_6H_5CHO in the reaction above and made 3.39 g of $C_6H_5CH_2OH$.

Calculate the percentage yield that this mass of $C_6H_5CH_2OH$ represents. [3]

Percentage yield = ... %

(iii) Cannizzaro's reaction is usually carried out using aqueous sodium hydroxide, rather than with water as shown in the equation above.

Write the equation for the Cannizzaro reaction of C_6H_5CHO with aqueous sodium hydroxide. [1]

....

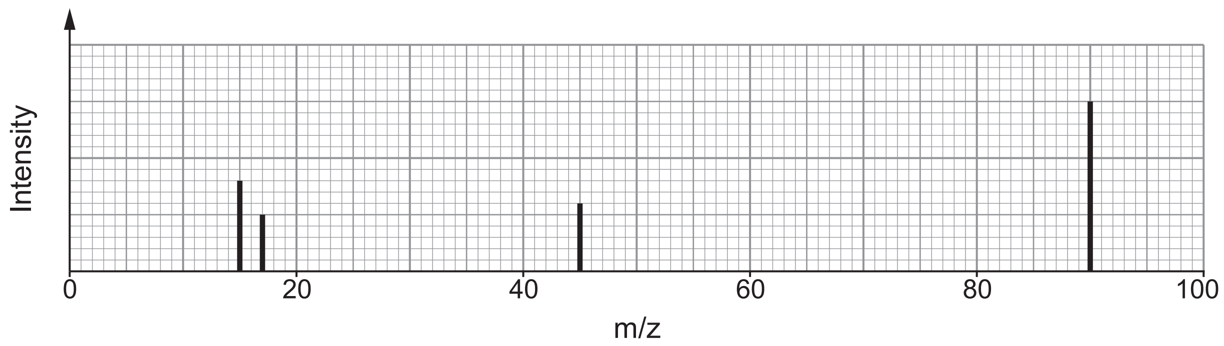
11

11. *(a)* Compound **X** contains only carbon, hydrogen and oxygen.

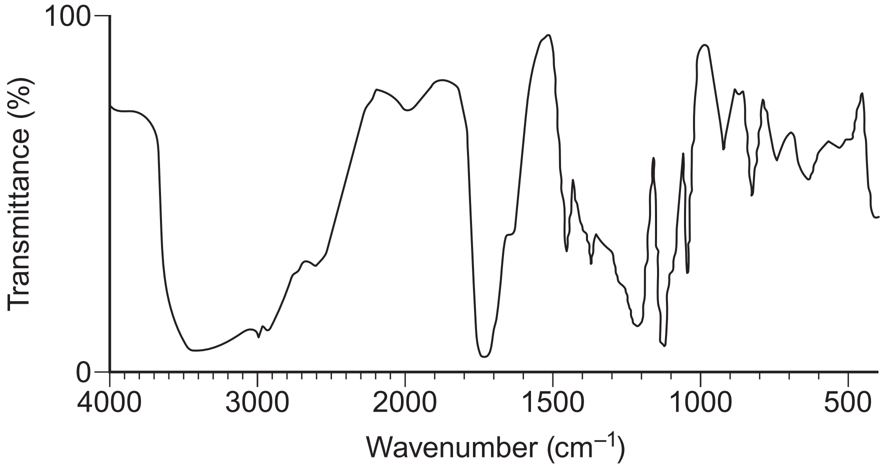
On analysis **X** was found to contain 40.0% carbon and 6.67% hydrogen by mass.

A simplified form of the mass spectrum, the IR spectrum and the low resolution ^{1}H NMR spectrum of **X** are shown.

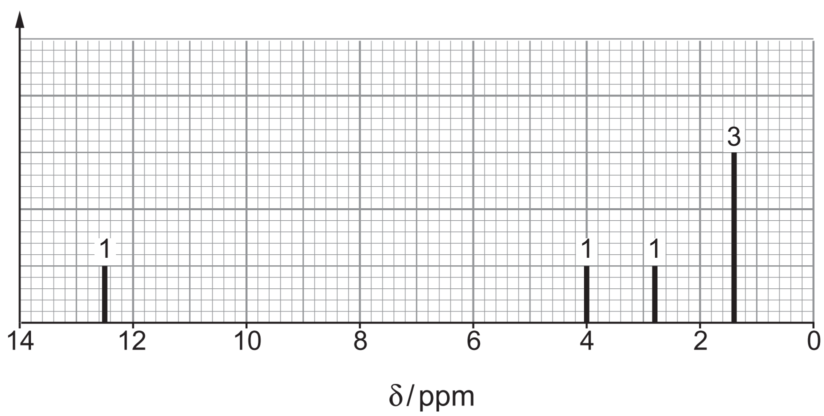
Mass spectrum



IR spectrum



^{1}H NMR spectrum



The numbers above the peaks show the relative areas of the peaks.

 (2410U20-1)

Examiner only

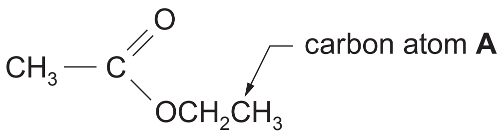
Use this information to identify compound **X**.

You must use information from **all** the sources given and explain how you used it. [10]

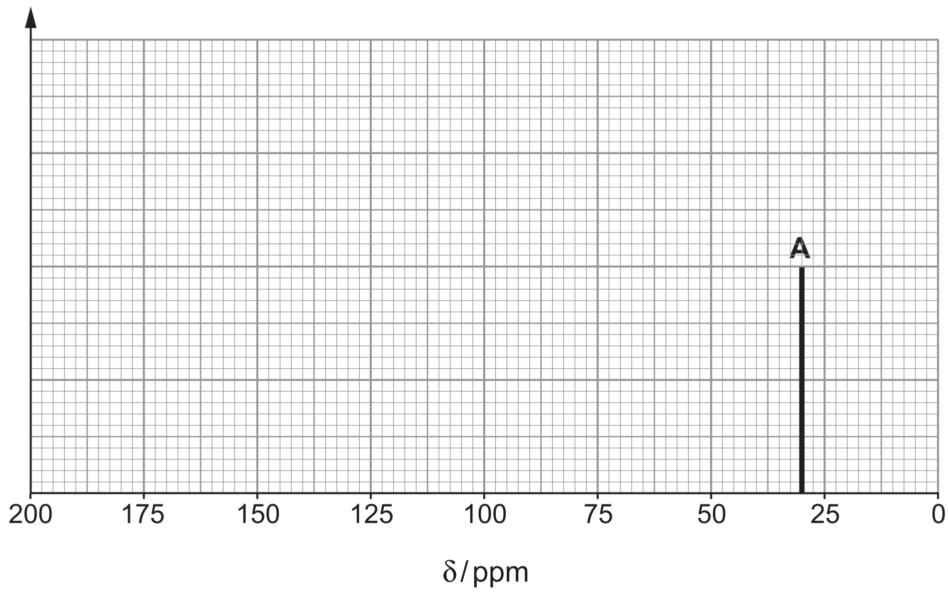
Examiner only

(b) Information about the structure of organic compounds can also be found using a ^{13}C NMR spectrum.

(i) On the axes below sketch the ^{13}C NMR spectrum that you would expect to obtain from ethyl ethanoate.



Label the diagram to show the species causing each peak.
The peak caused by carbon atom **A** is already included. [3]



(ii) A student said that valuable information about the nature of a sample being analysed could be obtained from the peak heights in both the ^{13}C and ^{1}H NMR spectra.

Comment on whether the student is correct. Give a reason for your answer. [1]

....

....

14

END OF PAPER

Surname	Centre Number	Candidate Number
First name(s)		2

GCE AS/A LEVEL

2410U20-1

Z22-2410U20-1

FRIDAY, 27 MAY 2022 – AFTERNOON

CHEMISTRY – AS unit 2
Energy, Rate and Chemistry of Carbon Compounds

1 hour 30 minutes

	For Examiner's use only		
	Question	**Maximum Mark**	**Mark Awarded**
Section A	**1. to 5.**	**10**	
Section B	**6.**	**15**	
	7.	**17**	
	8.	**10**	
	9.	**12**	
	10.	**16**	
	Total	**80**	

2410U201
01

ADDITIONAL MATERIALS

In addition to this examination paper, you will need a:
- calculator;
- **Data Booklet** supplied by WJEC.

INSTRUCTIONS TO CANDIDATES

Use black ink or black ball-point pen. Do not use gel pen or correction fluid. You may use a pencil for graphs and diagrams only.

Write your name, centre number and candidate number in the spaces at the top of this page.

Section A Answer **all** questions.
Section B Answer **all** questions.

Write your answers in the spaces provided in this booklet. If you run out of space, use the additional page(s) at the back of the booklet, taking care to number the question(s) correctly.

Candidates are advised to allocate their time appropriately between **Section A (10 marks)** and **Section B (70 marks)**.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets at the end of each question or part-question.

The maximum mark for this paper is 80.

Your answers must be relevant and must make full use of the information given to be awarded full marks for a question.

The assessment of the quality of extended response (QER) will take place in **Q9**(c).

JUN222410U20101

BE/DJ*(S22-2410U20-1)

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SECTION A

Answer **all** questions.

1. Bromine water can be used to test for alkenes.

(a) (i) State the expected colour change for a positive test for alkenes. [1]

....

(ii) Draw the structure of the product formed when propene reacts with bromine water. [1]

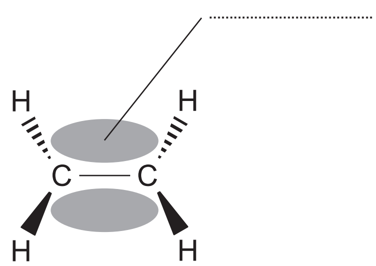
(b) Identify another reagent that can be used to test for the presence of alkenes. [1]

....

2. Bonds in hydrocarbons are formed by the overlap of orbitals between each atom.

(a) Draw an *s*-orbital and a *p*-orbital in the space below. [1]

(b) Name the type of bond shown in the diagram below. [1]



Examiner only

3. A student suspects an unlabelled organic liquid is a carboxylic acid. Name the reagent(s) that must be added to the unknown organic liquid to test for the presence of a carboxylic acid. Give the expected observations for a positive result. [2]

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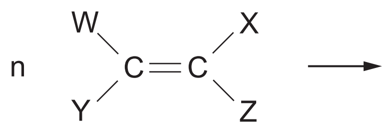
4. Complete the equation below to show the product of addition polymerisation. [1]



5. State the meaning of the term 'standard enthalpy change of formation'. [2]

....

....

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2410U201
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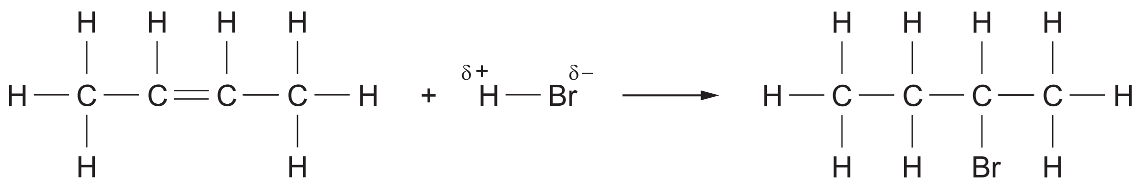
Examiner only

SECTION B

Answer **all** questions.

6. Butanone can be prepared from but-2-ene using a three-step synthesis.

(a) In the first step, but-2-ene is reacted with HBr to form 2-bromobutane.

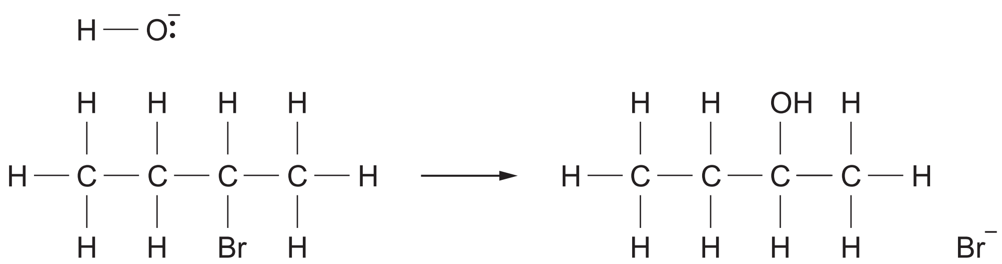


(i) Circle the species that represents the electrophile. [1]

(ii) Name the type of bond fission that takes place in the H—Br bond in the first step of the mechanism. [1]

....

(b) In the second step, 2-bromobutane undergoes nucleophilic substitution to form butan-2-ol.



(i) Use curly arrows to complete the equation to show the mechanism of the nucleophilic substitution. Include any relevant partial charges. [2]

(ii) Give the reagents and conditions required for this nucleophilic substitution. [2]

....

....

(iii) State the classification of alcohol to which butan-2-ol belongs. [1]

....

2410U201
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Examiner only

(c) In the final step, butan-2-ol is heated with acidified potassium manganate(VII) to produce butanone.

(i) State the role of the acidified potassium manganate(VII) in this reaction. [1]

....

(ii) Explain why butanone can be removed from the reaction as it is formed using distillation, leaving unreacted butan-2-ol in the reaction mixture. [2]

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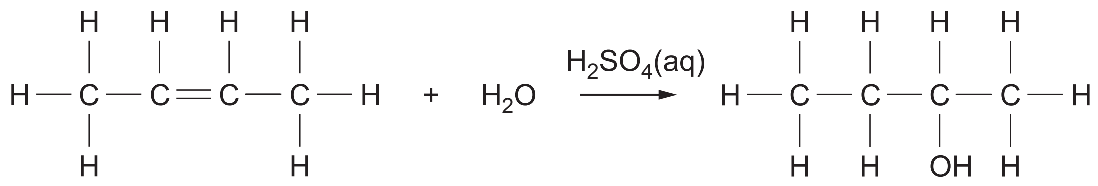
(d) Butan-2-ol can also be made directly by hydration of but-2-ene in the presence of dilute sulfuric acid, which acts as a catalyst.



(i) Suggest why the overall yield of the two-step synthesis is likely to be lower than the yield of the direct hydration. [1]

....

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Examiner only

(ii) Reaction of butan-2-ol with **concentrated** sulfuric acid results in the formation of three isomers with the formula C_4H_8.

I. Name the type of reaction used to form alkenes from alcohols. [1]

....

II. Give the structure and **name** of the three isomers formed. [3]

Structure	Structure	Structure
Name:	Name:	Name: ....

2410U201 07

15

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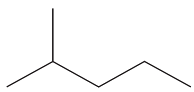
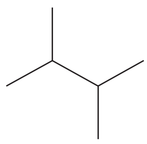
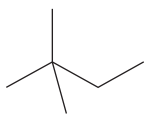
7. Petroleum ether (50–70) is a mixture of different alkanes extracted from crude oil which is commonly used as an organic solvent. The major components of petroleum ether (50–70) are the structural isomers of C_6H_{14}.

(a) (i) Give the meaning of the term 'structural isomer'. [1]

....

....

(ii) Complete the table below showing important information about the isomers of C_6H_{14}. [3]

Name	Shortened structural formula	Skeletal formula	Boiling temperature / °C
hexane	$CH_3CH_2CH_2CH_2CH_2CH_3$		69
2-methylpentane			62
3-methylpentane	$CH_3CH_2CH(CH_3)CH_2CH_3$		63
	$(CH_3)_2CHCH(CH_3)_2$		58
2,2-dimethylbutane	$CH_3C(CH_3)_2CH_2CH_3$		50

Examiner only

(iii) State the relationship between the boiling temperature and the carbon chain length. Explain this relationship in terms of intermolecular forces. [2]

....

....

....

....

(b) Hexane can be used as a fuel in a combustion reaction.

(i) Write an equation for the complete combustion of hexane in excess oxygen. [2]

....

(ii) The enthalpy change of combustion ($\Delta_c H^\theta$) for hexane is approximately $-4160\,kJ\,mol^{-1}$. Explain why the enthalpy change of combustion for the isomers of hexane should be similar. [2]

....

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....

(iii) 2,2-dimethylbutane is the isomer of C_6H_{14} which ignites most readily. Suggest a reason for this. [1]

....

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2410U201
09

Examiner only

(iv) When hexane burns in a limited supply of oxygen it undergoes a different reaction known as incomplete combustion:

$$C_6H_{14} \quad + \quad 6.5O_2 \longrightarrow 6CO \quad + \quad 7H_2O$$

The bond enthalpy values for the bonds present in these molecules are given below:

Bond	Average bond enthalpy / $kJ\,mol^{-1}$
C — C	348
C — H	413
O ═ O	495
C ≡ O (in CO)	1072
O — H	464

I. Using a Hess cycle or otherwise, calculate the enthalpy change of this reaction. [3]

enthalpy change = ... $kJ\,mol^{-1}$

II. **Use the enthalpy values** from parts (b)(ii) and (b)(iv) I. to explain quantitatively why it is important to maintain an excess of oxygen while burning hexane as a fuel. [2]

....

....

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Examiner only

III. State a health hazard associated with the incomplete combustion of hexane. [1]

....

....

17

2410U201
11

8. Compound **A** contains only carbon, hydrogen and an unknown halogen.

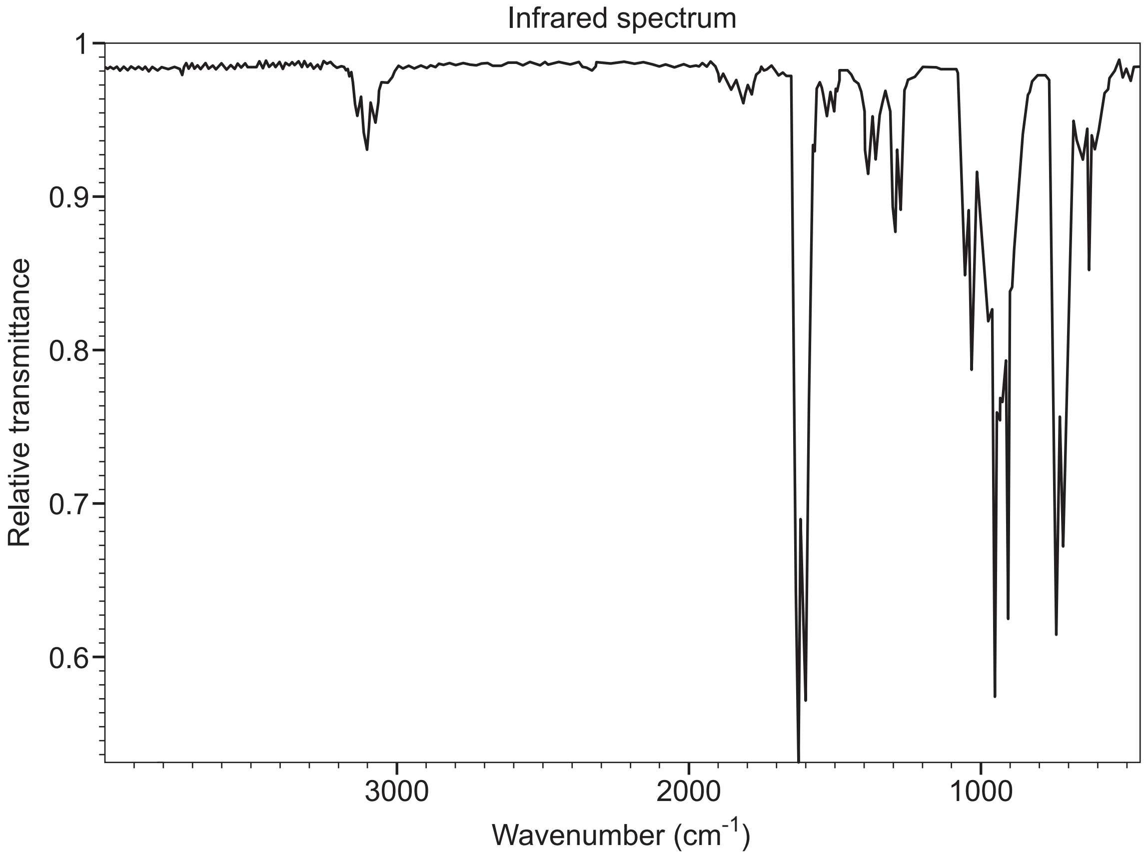
Refluxing compound **A** in aqueous sodium hydroxide followed by the addition of nitric acid and aqueous silver nitrate produces a white precipitate.

Elemental analysis of compound **A** indicates it contains 39.02% carbon and 3.25% hydrogen by mass.

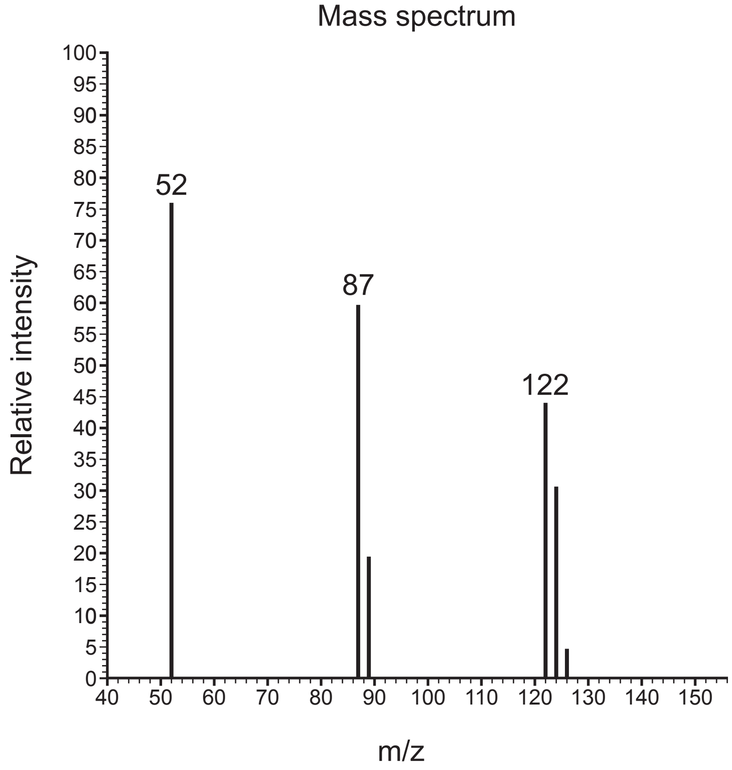
When bromine is added to compound **A**, 123 g of compound **A** reacts with 320 g of bromine.

The ^1H NMR spectrum of compound **A** consists of only one peak. The ^{13}C NMR spectrum of compound **A** consists of two peaks.

The infrared spectrum and simplified mass spectrum are shown below and overpage.



Examiner only



Use **all** of the data provided to determine the identity of compound **A**. Explain your answer using information from all of the data sources provided. [10]

Examiner only

10

Examiner only

9. Chloe was investigating the effect of using catalysts on the rate of reaction.

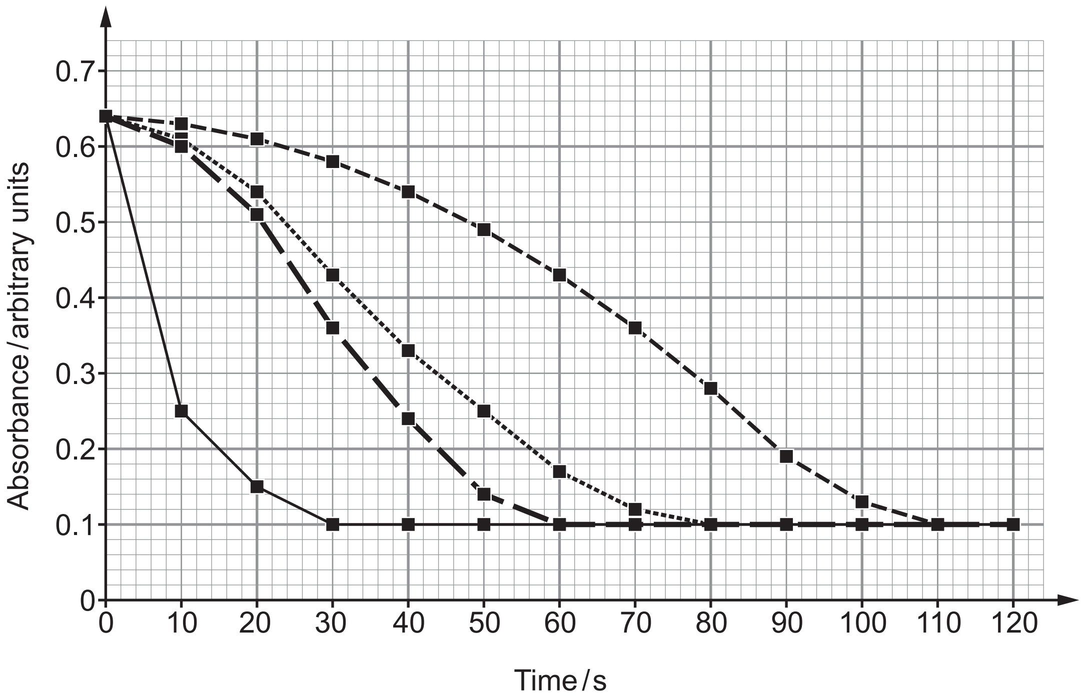
She added 50 cm^3 of 0.1 mol dm^{-3} iron(III) nitrate solution to 50 cm^3 of 0.2 mol dm^{-3} sodium thiosulfate solution. The reaction forms a deep violet iron(III) complex which is unstable and is gradually reduced to form a light green iron(II) complex.

Chloe monitored the rate of reaction by measuring the absorption of light at a wavelength of 500 nm every 10 seconds for two minutes using a data logger.

(a) The violet complex appears black at the beginning of the reaction. State the name of the technique used to monitor the rate of reaction by measuring the absorption of light. [1]

....

(b) Chloe repeated the experiment three times adding 1 cm^3 of a different catalyst each time at a concentration 0.10 mol dm^{-3}. Below is a graph showing her results:



Key: No catalyst; Fe^{2+}; Co^{2+}; Cu^{2+}

(i) State which catalyst is the most effective. Explain your answer. [2]

....

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Examiner only

(ii) Calculate the initial rate of reaction for the reaction catalysed by the copper(II) ions. [2]

rate = ... s^{-1}

(iii) Each catalysed reaction contained the same number of moles of catalyst at the beginning of the reaction. Calculate the moles of catalyst left at the end of the reaction. [1]

moles = ... mol

Examiner only

(c) Increasing temperature and the addition of a catalyst are two ways of increasing the rate of a reaction.

Using your knowledge of the Boltzmann distribution and particle theory, explain how the rate of reaction is increased using these two different methods.
You may use a diagram(s) to support your answer. [6 QER]

12

10. The crystallisation of sodium ethanoate from a super-saturated solution is used to release heat in reusable hand warmers.

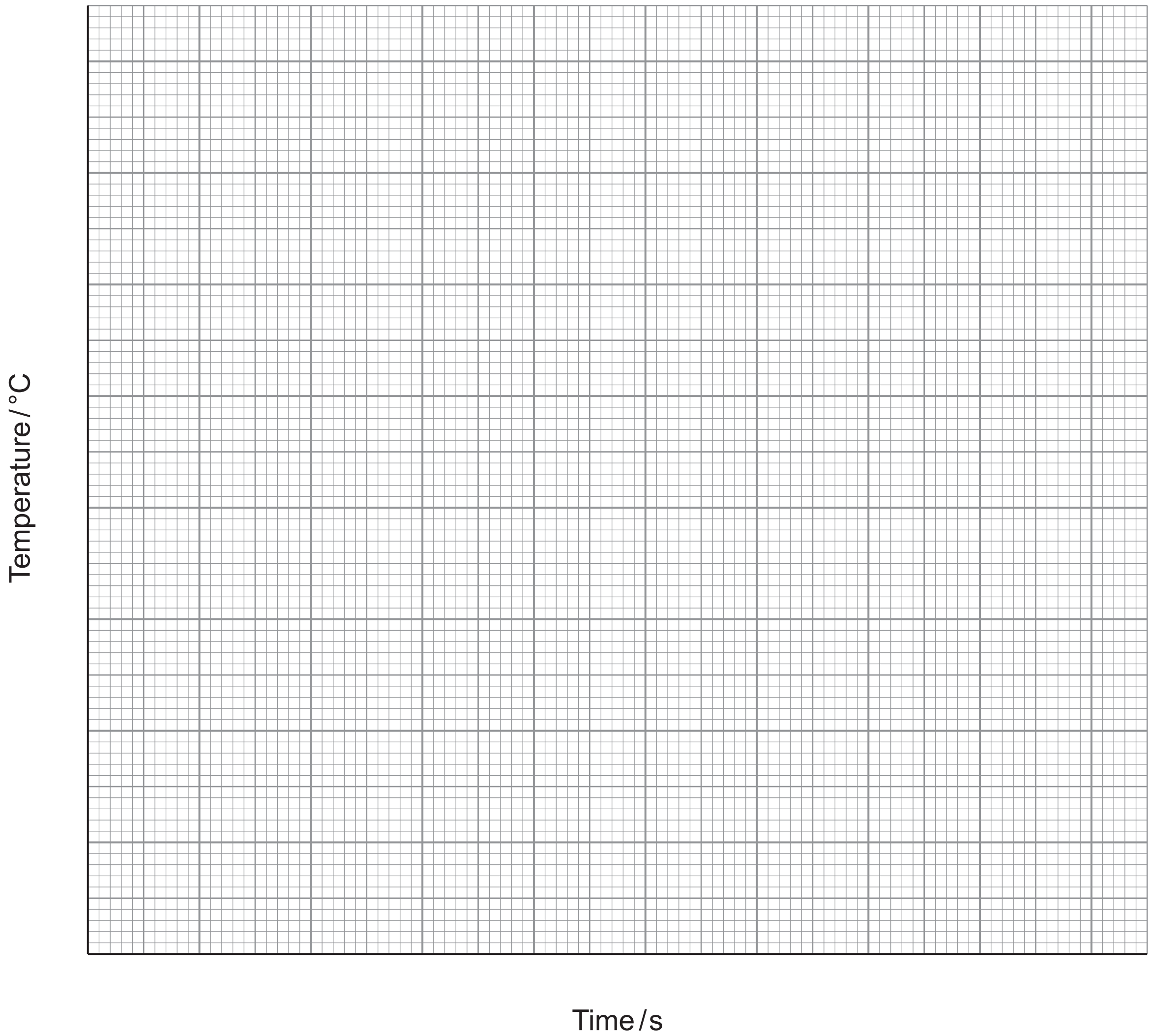
(a) A super-saturated solution of sodium ethanoate was made by dissolving 320 g of hydrated sodium ethanoate ($CH_3COONa.3H_2O$) in 60 cm^3 of hot water. It was then allowed to cool to room temperature, which was measured as 17 °C.

A thermometer was added to the solution, which caused the sodium ethanoate to start crystallising. The temperature of the process was recorded every 30 seconds for 3 minutes. The results are shown below:

Time / s	Temperature / °C
0	17
30	27
60	35
90	41
120	40
150	39
180	38

Examiner only

(i) Plot the results on the graph paper below. [2]



(ii) Use your graph to calculate the maximum temperature change for this crystallisation. [2]

maximum temperature change = ... °C

Examiner only

(iii) Use the **total mass** of the sodium ethanoate solution and the temperature change from the graph to calculate the enthalpy change of crystallisation per mole of sodium ethanoate. Assume the density of water is $1.00\,g\,cm^{-3}$ and the specific heat capacity of sodium ethanoate solution is $4.18\,J\,K^{-1}\,g^{-1}$.

$M_r(CH_3COONa.3H_2O) = 136$ [4]

enthalpy change = ... $kJ\,mol^{-1}$

(iv) Suggest a reason why the experimental enthalpy change is often lower than the theoretical enthalpy change. [1]

....

....

(b) Sodium ethanoate can be made in a neutralisation reaction. Complete the following equation: [2]

.... + ... $\longrightarrow$ CH_3COONa + CO_2 + H_2O

(c) The carboxylic acid used to produce sodium ethanoate can be produced using an oxidation reaction.

(i) Name the reagents and give the expected observations. [2]

....

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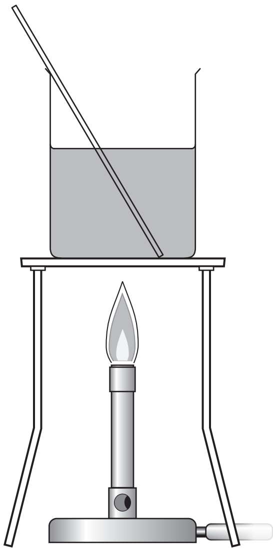
Examiner only

(ii) A student proposed that the apparatus below should be used to perform this oxidation reduction experiment.



The teacher said that this would not work and would be unsafe. Draw a labelled diagram of the apparatus that should be used in this experiment. [3]

16

END OF PAPER

Surname	Centre Number	Candidate Number
First name(s)		2

GCE AS/A LEVEL

2410U20-1

S23-2410U20-1

THURSDAY, 25 MAY 2023 – MORNING

CHEMISTRY – AS unit 2
Energy, Rate and Chemistry of Carbon Compounds

1 hour 30 minutes

	For Examiner's use only		
	Question	**Maximum Mark**	**Mark Awarded**
Section A	**1. to 7.**	**10**	
Section B	**8.**	**6**	
	9.	**6**	
	10.	**18**	
	11.	**10**	
	12.	**13**	
	13.	**17**	
	Total	**80**	

2410U201
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ADDITIONAL MATERIALS

In addition to this examination paper, you will need a:
- calculator;
- **Data Booklet** supplied by WJEC.

INSTRUCTIONS TO CANDIDATES

Use black ink or black ball-point pen.
Do not use gel pen or correction fluid.

You may use pencil for graphs and diagrams only.

Write your name, centre number and candidate number in the spaces at the top of this page.

Section A Answer **all** questions.
Section B Answer **all** questions.

Write your answers in the spaces provided in this booklet. If you run out of space, use the additional page(s) at the back of the booklet, taking care to number the question(s) correctly.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets at the end of each question or part-question.

The maximum mark for this paper is 80.

Your answers must be relevant and must make full use of the information given to be awarded full marks for a question.

The assessment of the quality of extended response (QER) will take place in **Q.8**.

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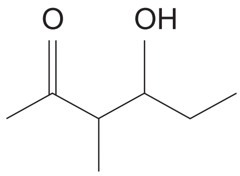
Examiner only

SECTION A

Answer **all** questions.

1. Give the molecular formula of the compound shown. [1]



Molecular formula ....

2. When fuels are burned with insufficient oxygen, incomplete combustion occurs and carbon monoxide is formed instead of carbon dioxide.

Write the equation for the incomplete combustion of propane, C_3H_8. Assume that carbon monoxide is the only carbon-containing product. [1]

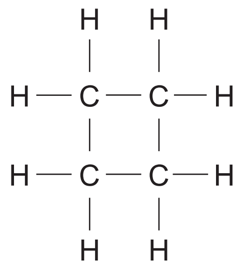
....

3. Name the reagent that is needed to change unsaturated oils into saturated fats. [1]

....

Examiner only

4. Cyclobutane is a cyclic hydrocarbon whose structure is shown.



Draw the structures of **two** structural isomers of this compound. [2]

2410U201 03

5. Propene reacts with hydrogen bromide.

$$C_3H_6 \quad + \quad HBr \longrightarrow C_3H_7Br$$

Draw the structures of the **two** isomers formed and give the name of the major product. [2]

Major product ....

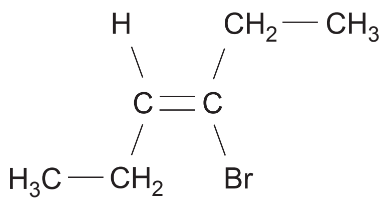
Examiner only

6. In the forward direction, the enthalpy change of reaction, ΔH, for a reversible endothermic reaction, has the numerical value of $60\,kJ\,mol^{-1}$.

If the activation energy, E_a, of the forward reaction is $100\,kJ\,mol^{-1}$, calculate the activation energy of the backward reaction. [1]

E_a = ... $kJ\,mol^{-1}$

7. Name the following *E-Z* isomer. [2]



....

10

Examiner only

SECTION B

Answer **all** questions.

8. The use of biofuels has been proposed as the way to ensure the sustainability of energy production in the future.

Discuss the extent to which you agree with this proposal, giving both advantages and disadvantages of the use of biofuels. Include relevant examples in your answer. [6 QER]

2410U201
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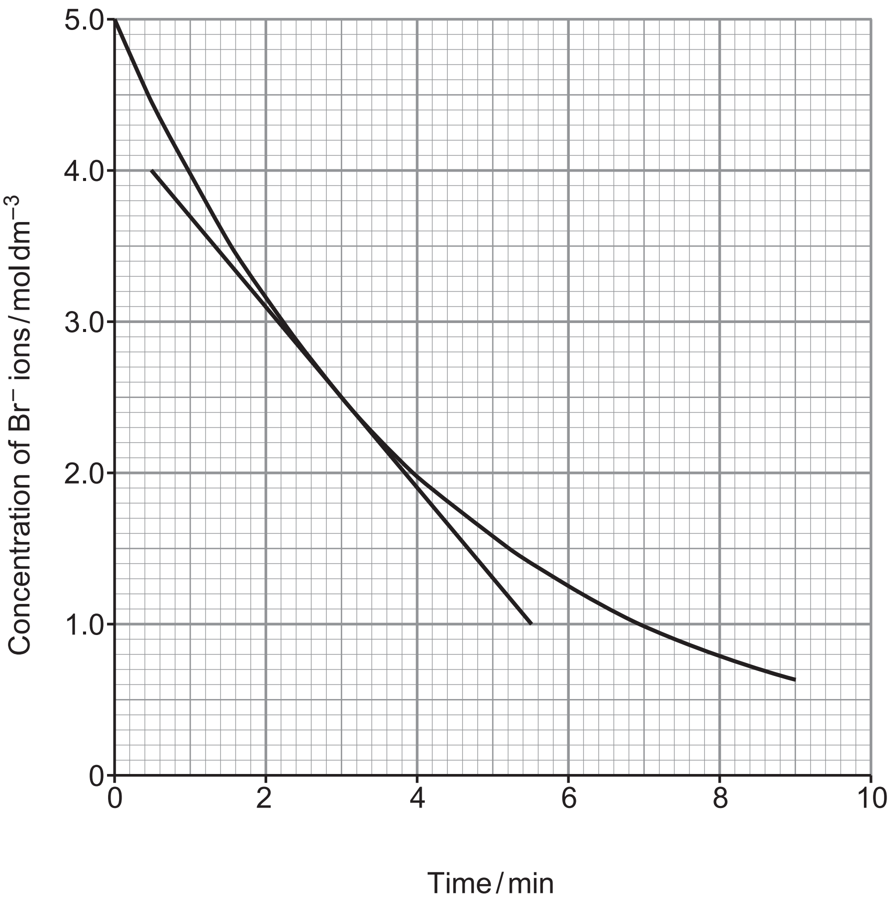
6

Examiner only

9. Bromide ions react with bromate(V) ions, BrO_3^-, as shown in the equation.

$$5Br^-(aq) \quad + \quad BrO_3^-(aq) \quad + \quad 6H^+(aq) \longrightarrow 3Br_2(aq) \quad + \quad 3H_2O(l)$$

Some students investigated the rate of this reaction by following the concentration of bromide ions over time. They plotted their results as shown.



Examiner only

(a) (i) The students calculated that the initial rate of the reaction was $1.20\,mol\,dm^{-3}\,min^{-1}$.

Use the tangent to the curve to calculate the rate of the reaction when the concentration of bromide ions had fallen to $2.50\,mol\,dm^{-3}$.

You **must** show your working. [2]

Rate = ... $mol\,dm^{-3}\,min^{-1}$

(ii) Use your answer to part (i) to suggest the relationship between the rate of this reaction and the concentration of bromide ions. [1]

....

....

(b) Suggest how the students could follow the rate of this reaction. Explain your answer. [2]

....

....

....

(c) In the experiment, the concentration of bromide ions fell from $5.0\,mol\,dm^{-3}$ to $2.0\,mol\,dm^{-3}$ over the first 4 minutes. The initial concentration of bromate(V) ions was $1.0\,mol\,dm^{-3}$.

$$5Br^-(aq) + BrO_3^-(aq) + 6H^+(aq) \longrightarrow 3Br_2(aq) + 3H_2O(l)$$

Using the equation, deduce the concentration of bromate(V) ions after 4 minutes. [1]

Concentration of bromate(V) ions = ... $mol\,dm^{-3}$

6

2410U201 07

Turn over.

10. (a) The enthalpy of neutralisation of an acid is defined as the enthalpy change when 1 mol of aqueous H^+ ions is neutralised by aqueous OH^- ions according to the equation shown. The reaction is exothermic.

$$H^+(aq) + OH^-(aq) \longrightarrow H_2O(l)$$

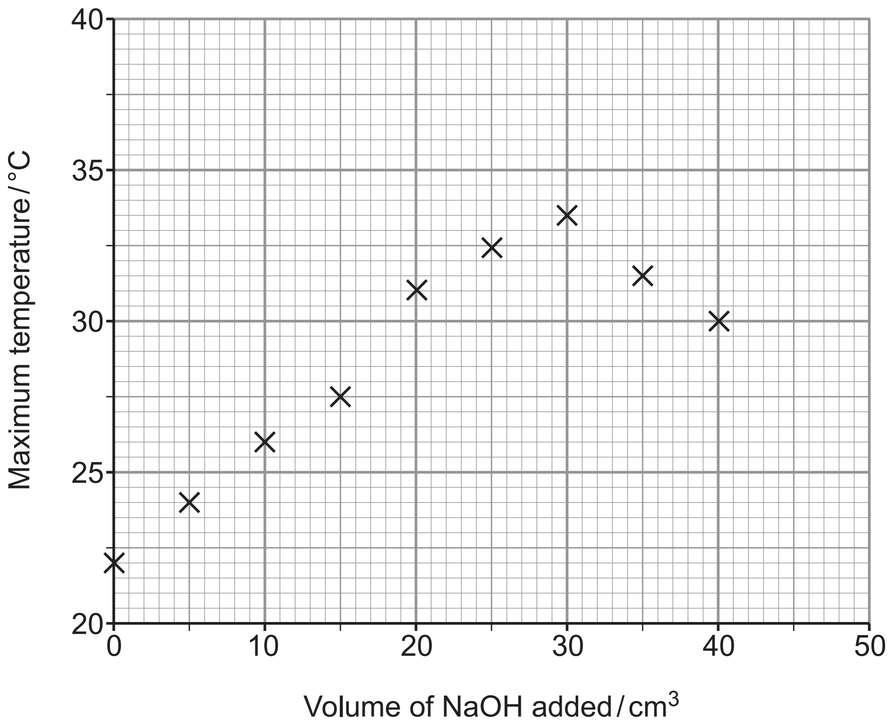
Some students followed the instructions below to determine the enthalpy change of neutralisation of methanoic acid, HCOOH.

1. Weigh 24.7 g of methanoic acid and mix with water to make 250 cm^3 of solution. Record the temperature of the solution.
2. Transfer eight 25.0 cm^3 portions of this solution into eight insulated cups.
3. Using a burette add 5.0 cm^3 of aqueous sodium hydroxide to the solution in the first cup. Stir and record the maximum temperature reached.
4. Add the following volumes of aqueous sodium hydroxide to each of the remaining cups in turn:

   10.0 cm^3 15.0 cm^3 20.0 cm^3 25.0 cm^3 30.0 cm^3 35.0 cm^3 40.0 cm^3

   Stir and record the maximum temperature reached in each cup.
5. Plot a graph of maximum temperature reached against volume of sodium hydroxide added.

Their results are plotted in the graph.



Examiner only

(i) Name the apparatus used to transfer exactly 25.0 cm^3 of methanoic acid solution into the insulated cups. [1]

....

(ii) State why a higher maximum temperature is recorded when increasing volumes of sodium hydroxide are added. [1]

....

....

(iii) Explain why the maximum temperature recorded decreases when more than 30 cm^3 of sodium hydroxide is added. [2]

....

....

....

(iv) On the graph, draw **one** straight line through the points that show an increase in maximum temperature and **another** straight line through the points that show a decrease in maximum temperature. [1]

(v) From the graph, deduce the volume of sodium hydroxide needed to neutralise 25.0 cm^3 of the methanoic acid solution and the temperature increase at that point.

Assume that the initial temperature of every 25.0 cm^3 of methanoic acid solution is 22.0 °C. [2]

Volume = ... cm^3

Temperature increase = ... °C

Examiner only

(vi) Use your answers to part (v) to calculate the amount of heat released by the neutralisation reaction. [2]

Heat released = ... J

(vii) Calculate the number of moles of methanoic acid in $25.0\,cm^3$ of the solution and hence the enthalpy change of neutralisation of methanoic acid. [3]

Enthalpy change of neutralisation = ... $kJ\,mol^{-1}$

Examiner only

(b) The experiment is repeated using hydrochloric acid instead of methanoic acid and a more negative value of the enthalpy change of neutralisation is calculated.

Suggest and explain a reason for this difference. [2]

....

....

....

(c) (i) Write the equation for the reaction that occurs when solid copper(II) carbonate is added to aqueous methanoic acid to form aqueous copper(II) methanoate, $(HCOO)_2Cu$.

Include state symbols. [2]

....

(ii) State what is observed during the reaction in part (i). [2]

....

....

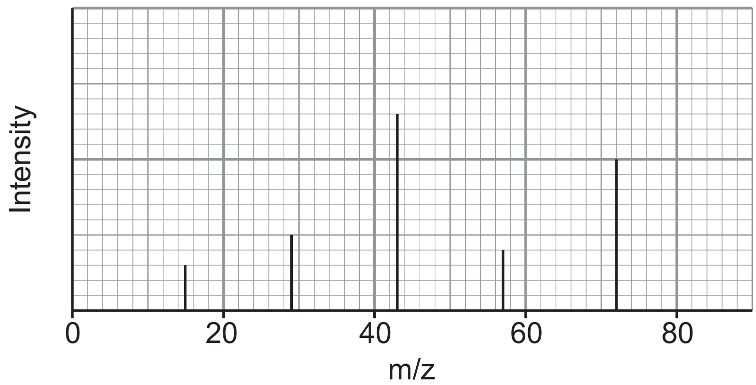
2410U201
11

18

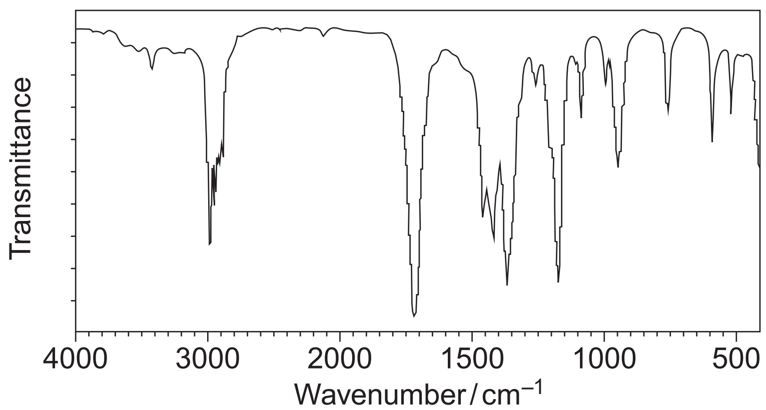
11. (a) Compound **X** contains carbon, hydrogen and oxygen only. It has no reaction with acidified potassium dichromate.

Simplified versions of its mass spectrum, IR spectrum and ^{13}C NMR spectrum are shown.

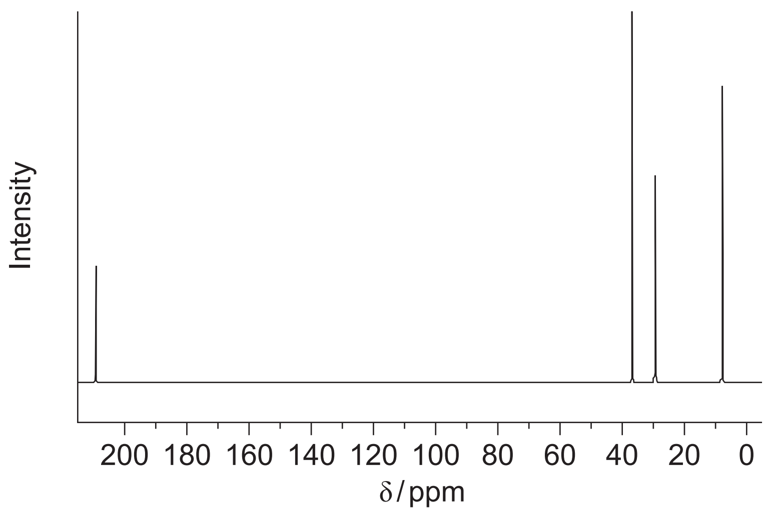
Mass spectrum



IR spectrum



^{13}C NMR spectrum



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Examiner only

Identify compound **X**.

You **must** use information from **all** the sources given and explain how you used it. [8]

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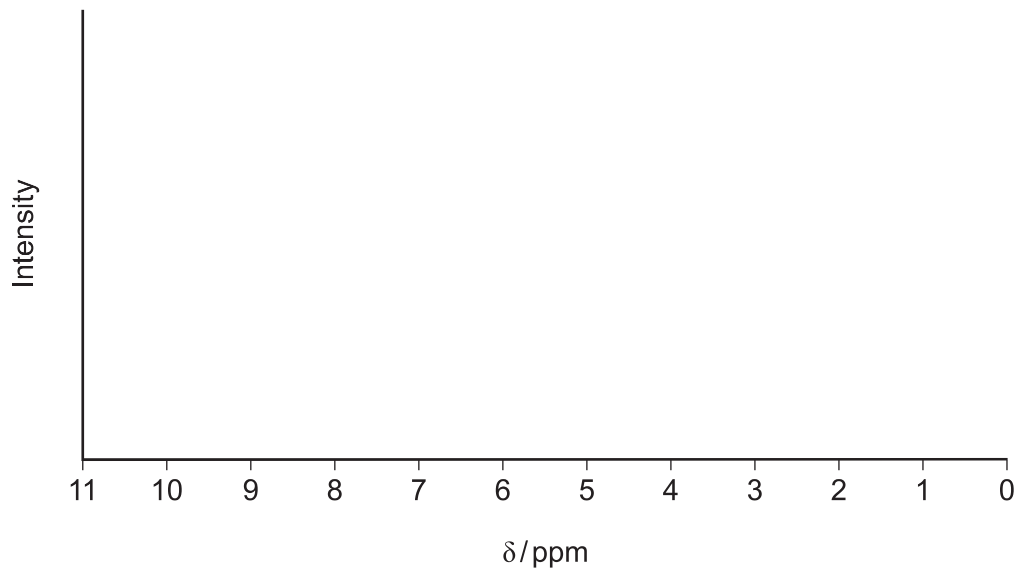
Compound **X**

Examiner only

(b) On the axes below, complete the low resolution ^{1}H NMR spectrum you would expect for the compound you identified in part (a).

You should indicate where you would expect to see peaks **and** the relative intensities of the peaks. [2]



10

Examiner only

12. (a) **Y** is a halogenocompound in which each molecule contains one atom of chlorine, bromine or iodine.

(i) Describe a chemical test to determine which halogen is present in **Y**. [3]

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(ii) **Y** contains four carbon atoms in each molecule. The percentage by mass of halogen present in **Y** is less than 40%.

Identify **Y**. Explain how you reached your conclusion. [2]

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Examiner only

(b) (i) Halogenoalkanes react with aqueous sodium hydroxide.

Draw the mechanism to show the reaction of 1-chloropropane with aqueous sodium hydroxide.

You should include all charges, partial charges and lone pairs, and curly arrows to show electron movement. [4]

(ii) Name the type of reaction shown in part (i). [1]

....

(c) Halogenoalkanes can also take part in an elimination reaction.

2-Chloropentane undergoes elimination in a similar way to 1-chloropropane.

(i) Give the reagent and conditions needed for 2-chloropentane to undergo elimination. [1]

....

(ii) When 2-chloropentane undergoes elimination, two structural isomers are formed.

Give the structures of these **two** isomers. [2]

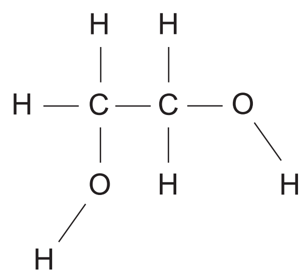
13

Examiner only

13. (a) In cold weather the wings of an aeroplane can become covered in ice. For safety reasons the wings must be de-iced.

The liquid ethane-1,2-diol, CH_2OHCH_2OH, is used, mixed with water, since this lowers the freezing temperature of water and causes the ice to melt.

(i) Use the diagram below to show the intermolecular forces that allow ethane-1,2-diol to dissolve in water. [3]



(ii) Suggest a reason why the addition of ethane-1,2-diol lowers the freezing temperature of water. [1]

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Examiner only

(b) Ethane-1,2-diol can be oxidised to ethanedioic acid, $(COOH)_2$.

(i) Suggest an oxidising agent suitable to carry out this reaction. [1]

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(ii) Write the equation for this reaction. Use [O] for the oxidising agent. [2]

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(iii) To ensure complete oxidation the reagents are refluxed.

Draw and label the apparatus as it is being used in this reaction. [2]

(iv) A sample of the reacting mixture was taken during the reflux process and a mass spectrum was produced. One of the peaks recorded was at m/z 58.

Suggest the identity of the **molecular ion** that gave this peak. [1]

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(v) The reaction can be used to prepare a sample of solid ethanedioic acid. This is generally hydrated as $(COOH)_2.xH_2O$ where x is an integer.

2.00 g of ethane-1,2-diol was oxidised and 3.94 g of hydrated ethanedioic acid was produced.

Calculate the relative molecular mass of hydrated ethanedioic acid and hence the value of x in its formula. [4]

x = ...

Examiner only

(vi) I. Complete the equation for the reaction which occurs when ethanedioic acid is heated with excess methanol in acidic conditions.

Clearly show the structure of the organic product. [2]

$$\begin{array}{c} COOH \\ | \\ COOH \end{array} \quad + \quad 2CH_3OH \longrightarrow \qquad\qquad\qquad + $$

II. Name the type of functional group present in the organic product. [1]

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17

END OF PAPER